Name:___________

Double Award Science – Applied Unit 1 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /75	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	18	20	24	28	33	38	43
2018	9	12	17	22	28	34	40
2019	7	10	16	22	28	34	40
2022	4	6	13	20	28	34	40
2023	8	11	16	21	26	31	39

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3445UA0-1

S17-3445UA0-1

APPLIED SCIENCE (Double Award)

Unit 1: Energy, Resources and the Environment
HIGHER TIER

WEDNESDAY, 14 JUNE 2017 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**19**	
2.	**12**	
3.	**9**	
4.	**6**	
5.	**13**	
6.	**16**	
Total	**75**	

3445UA01 01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

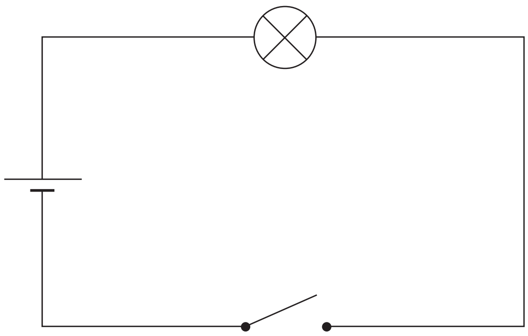
INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 5*(b)* is a quality of extended response (QER) question where your writing skills will be assessed.
You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
A periodic table is printed on page 16.

Examiner only

Answer **all** *the questions in the spaces provided.*

1. Students are investigating how the current through a lamp varies with the voltage across it.



(a) (i) Part of the circuit used is shown above. **Add** a variable resistor, ammeter and voltmeter to the circuit diagram. [3]

(ii) Describe the method you would use to take a range of results. [3]

....

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....

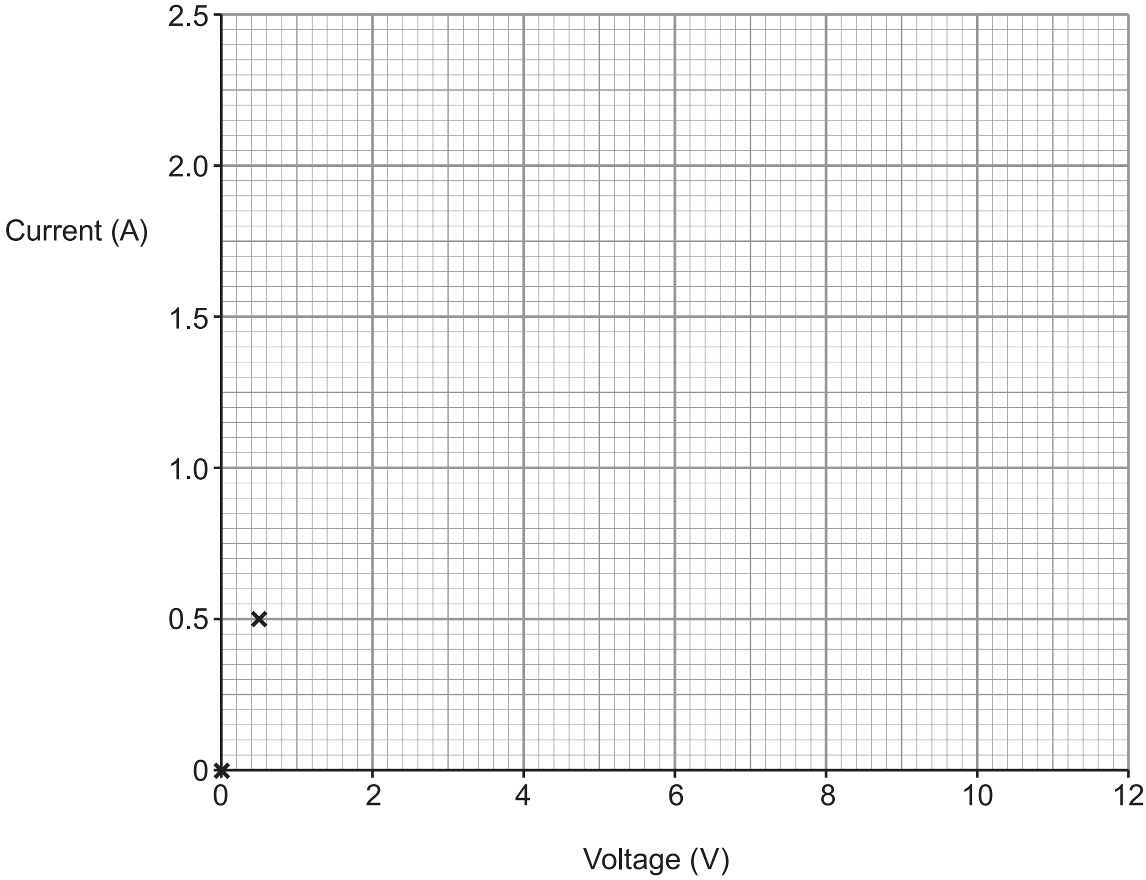
(b) Results from the experiment are shown below.

Voltage (V)	Current (A)
0.0	0.0
0.5	0.5
2.0	0.8
4.0	1.1
6.0	1.4
8.0	1.6
10.0	1.8
12.0	1.9

Examiner only

(i) Plot the data on the grid below and draw a suitable line. [3]

The first two points have been plotted for you.



(ii) Describe the relationship between current and voltage for the lamp. [2]

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....

....

(c) (i) Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the lamp at 0.5 V. [2]

resistance = ... Ω

3445UA01 03

Examiner only

(ii) State how the resistance of the lamp varies between 0.5 V and 12 V. [1]

....

(d) (i) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the lamp at 0.5 V. [2]

power = ... W

(ii) State how the power of the lamp varies between 0.5 V and 12 V. [1]

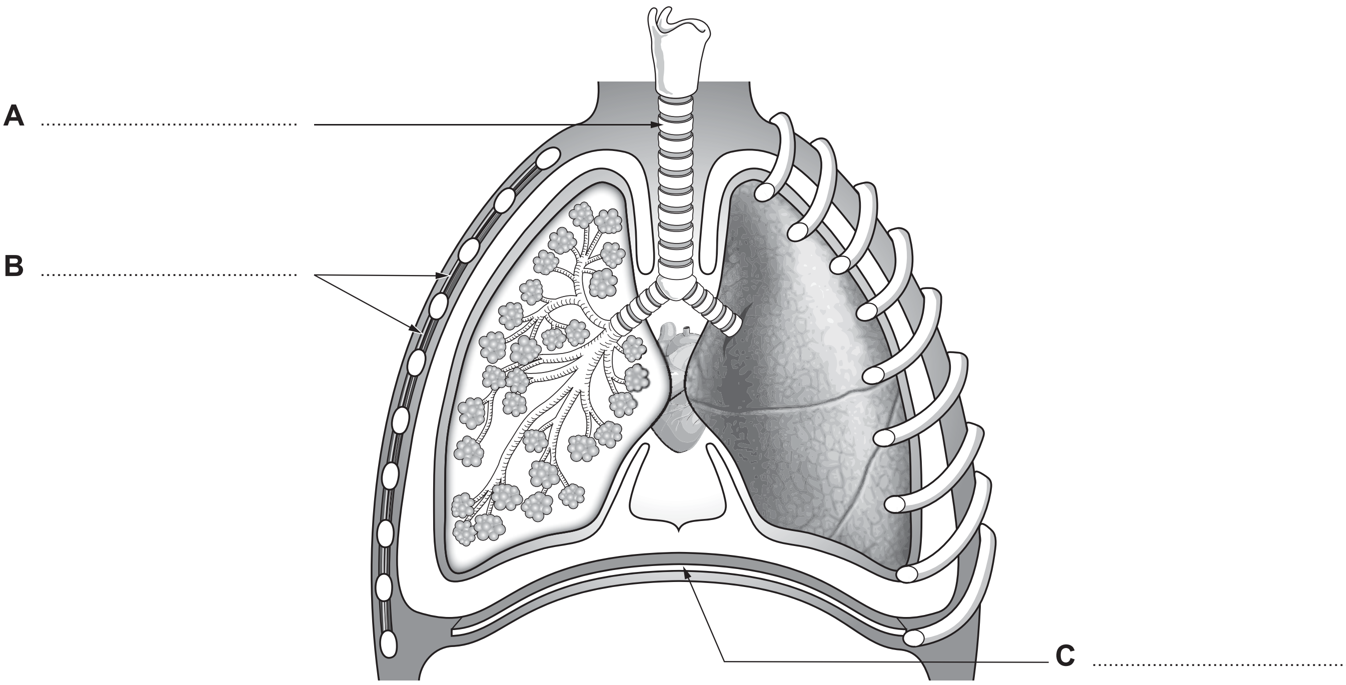
....

(e) The experiment was repeated with a length of wire at constant temperature. The current through the wire was 1.5 A when the voltage across it was 12 V.

Add a line to your graph to show how current varies with voltage for this wire. [2]

19

2. Sports science students are learning about the human respiratory system.



(a) (i) Label the diagram. [3]

(ii) Explain how changes to **B** and **C** allow us to breathe in. [3]

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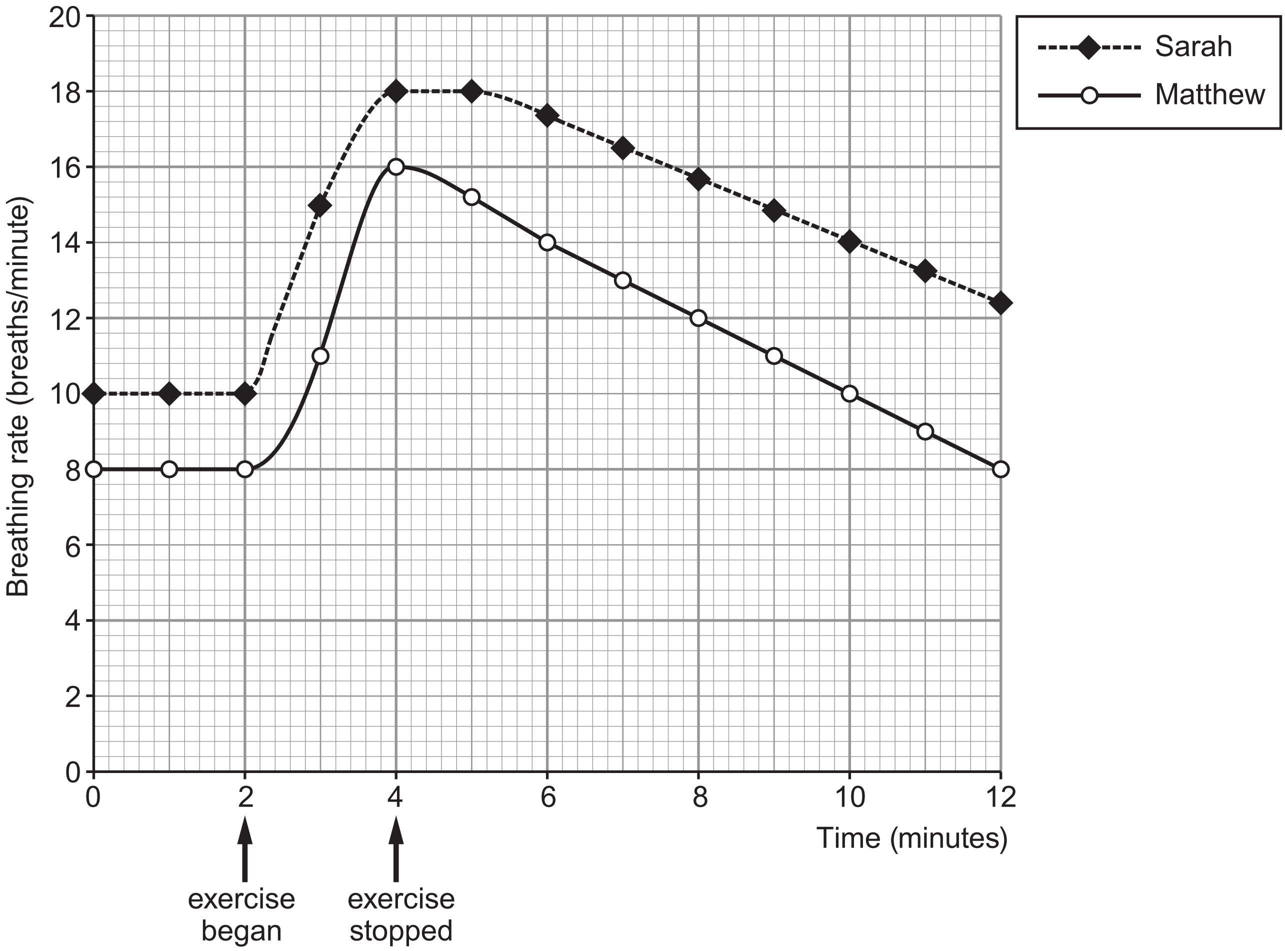
Examiner only

(b) Two students, Matthew and Sarah, studied the effect of exercise on breathing rate. They followed the method below:

1. record breathing rate before exercising
2. exercise for 2 minutes by running on a treadmill
3. measure the breathing rates during and after exercise up until 12 minutes

Their results are shown on the graph below.



Compare the changes in breathing rate of Sarah and Matthew between 8 and 12 minutes. [3]

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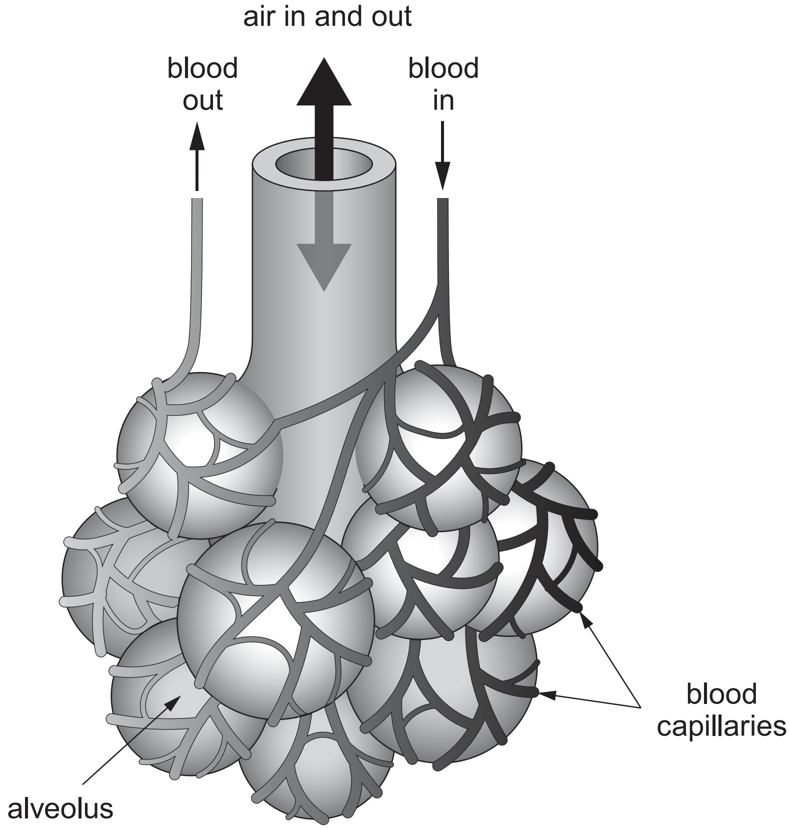
Examiner only

(c) Explain the process of gas exchange between the blood and the air inside the alveoli. [3]



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3445UA01 07

12

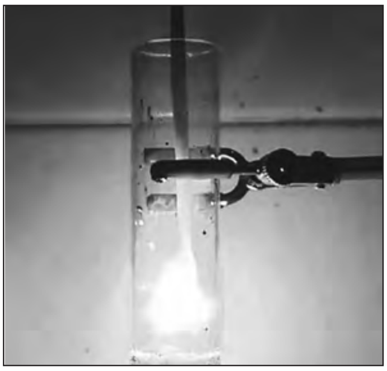
Examiner only

3. Magnesium oxide has various applications in water treatment. It is an effective base, offering a safe method of changing pH compared to other neutralising agents such as caustic soda. It also helps prevent scale formation in boilers, heat exchangers, and piping.

(a) State **one** adverse effect of scale on household plumbing. [1]

....

(b) Burning magnesium in a gas jar of oxygen produces magnesium oxide on a small scale.



(i) Describe the atomic structure of magnesium. [3]

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(ii) Draw a diagram to show the electronic structure of a magnesium atom. [1]

(iii) Write a balanced symbol equation for the reaction between magnesium and oxygen. [3]

....

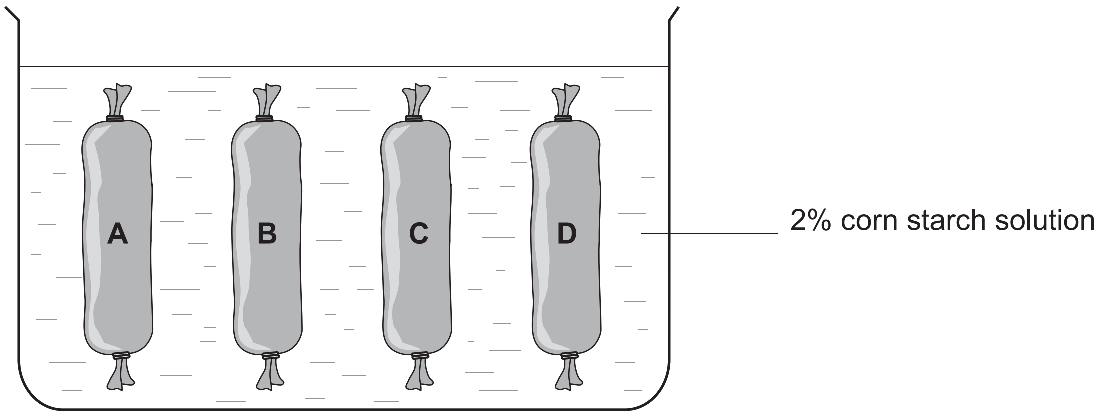
(iv) State how the electron configuration of a magnesium atom changes when magnesium reacts with oxygen. [1]

....

9

Examiner only

4. Starch is usually insoluble in water, but some types of starch e.g. corn starch can be treated to dissolve in water. Four pieces of Visking tubing **A**, **B**, **C** and **D**, contain equal volumes of corn starch solution of different concentrations. They are placed in a beaker containing a 2% corn starch solution and left for 45 minutes.



(a) Describe, in terms of molecules, the structure of starch. [2]

....

....

(b) After 45 minutes:

- the volume of **A** is unchanged
- the volume of **B** has increased
- the volume of **C** has decreased
- the volume of **D** has increased more than **B**

(i) Explain which tube contained a corn starch solution of lower concentration than that in the beaker. [2]

....

....

(ii) Explain why tube **A** must have contained the 2% corn starch solution. [2]

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6

Examiner only

5. The periodic table of elements is ordered by atomic number and electron configuration.

The tables below show some properties of alkali metals (group 1) and halogens (group 7).

Alkali metal	Atomic number	Melting point (°C)	Boiling point (°C)	Density at 20°C (g/cm^3)	Radius of the atom (units)
lithium	3	180.5	1 342	0.534	1.67
sodium	11	97.7	883	0.971	....
potassium	19	....	759	0.862	2.43
rubidium	37	39.3	688	1.532	2.65
caesium	55	28.4	671	1.873	2.98
francium	87	27.0	677		

Halogen	Atomic number	Melting point (°C)	Boiling point (°C)	Density at 20°C (g/cm^3)	Radius of the atom (units)
fluorine	9	–219	–188	0.0017	0.64
chlorine	17	....	–34	0.0032	0.99
bromine	35	–7	59	3.1	1.14
iodine	53	114	184	4.9	1.33
astatine	85	302	380	7.0	1.40

Use the information in the tables and your knowledge to answer the following questions.

(a) **Complete** the tables by estimating the missing values. [3]

(b) Compare the properties of the alkali metals and their trends with those of the halogens. [6 QER]

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Examiner only

(c) Explain why the reactivity of halogens changes from the top to the bottom of the group. [3]

(d) **Tick** (✓) the box next to the pair of elements that would react **most** violently. [1]

chlorine and lithium ☐

fluorine and lithium ☐

chlorine and sodium ☐

fluorine and sodium ☐

13

Examiner only

6. Rising energy prices are affecting many households. The government can help households keep their energy bills as low as possible. Domestic buildings account for about 35% of final UK energy consumption. Of this, heating rooms is responsible for about 60% and heating hot water accounts for about 15%. The UK is committed to reduce the CO_2 emissions caused by domestic buildings by 80% by 2050.

Statistics for a typical house with no insulation are shown below. The table also shows government estimates of the current usage of each method.

Part of house	Heat loss from uninsulated house (%)	Method of reducing heat loss	Cost (£)	Annual saving on energy bills (£)	% of homes already using this method
roof	25	loft insulation	450	150	91
windows & doors	10	double glazing	3900	90	70
walls	40	cavity wall insulation	1200	250	42
floor	15	carpets	800	40	90
draughts	10	draught excluders	60	5	85

At present, of the 95% of homes with boilers, 800000 have no controls at all and over 70% lack a timer, sufficient hot water tank insulation, room thermostat and thermostatic radiator valves. It is estimated that upgrading all homes so they have these four features could save 30% of the energy used for heating and hot water.

(a) A heating engineer estimates that to maintain the uninsulated house with no heating controls at 20°C will cost £1800 in energy bills per year.

Use the information above to answer the following questions.

(i) Calculate how much of this energy cost is lost through the roof. [2]

cost = £ ....

(ii) The engineer recommends that loft insulation should be installed first.

Explain why this is the best advice even though it will not make the biggest saving in energy bills. [2]

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Examiner only

(iii) Explain why the engineer can only estimate the cost of the energy bills. [3]

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(iv) After fitting a timer, new hot water tank insulation, room thermostat and thermostatic radiator valves a householder's energy bills would reduce to £1 300. Explain whether this is consistent with the estimated saving of 30% of the energy used. [4]

....

....

(v) Use the equation:

$$\text{units used} = \frac{\text{total cost}}{\text{cost of one unit}}$$

to calculate the decrease in units used if the savings in part (iv) above were made. [3]

Assume each unit costs 18 p.

Decrease in units used = ... kWh

(b) Explain how reducing household energy loss will benefit the environment. [2]

....

....

....

16

END OF PAPER

THE PERIODIC TABLE

1	2					Group						3	4	5	6	7	0
							1 H Hydrogen 1										4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3445UA0-1

S18-3445UA0-1

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

HIGHER TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	9	
3.	13	
4.	13	
5.	9	
6.	11	
7.	10	
Total	75	

3445UA01
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 7*(a)* is a quality of extended response (QER) question where your writing skills will be assessed.
You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
A periodic table is printed on page 20.

Examiner only

Answer **all** *questions.*

1. Students are making the fertiliser potassium nitrate in the laboratory. Most fertilisers are formed by the reaction of an acid and an alkali.

(a) Complete the word equation for the formation of potassium nitrate. [2]

.... + potassium hydroxide $\longrightarrow$ potassium nitrate + ...

(b) To make the fertiliser, it is important that the students are able to exactly neutralise the acid with alkali. To do this the students carefully measured 25 cm^3 of acid into a flask and titrated just enough alkali to neutralise the acid.

The results of the experiment are shown below.

Trial	Volume of alkali added at endpoint (titre) (cm^3)
1	20.5
2	20.8
3	18.5
4	20.0
5	19.9

(i) The reading for trial 3 was considered to be an anomalous result. Give a reason why. [1]

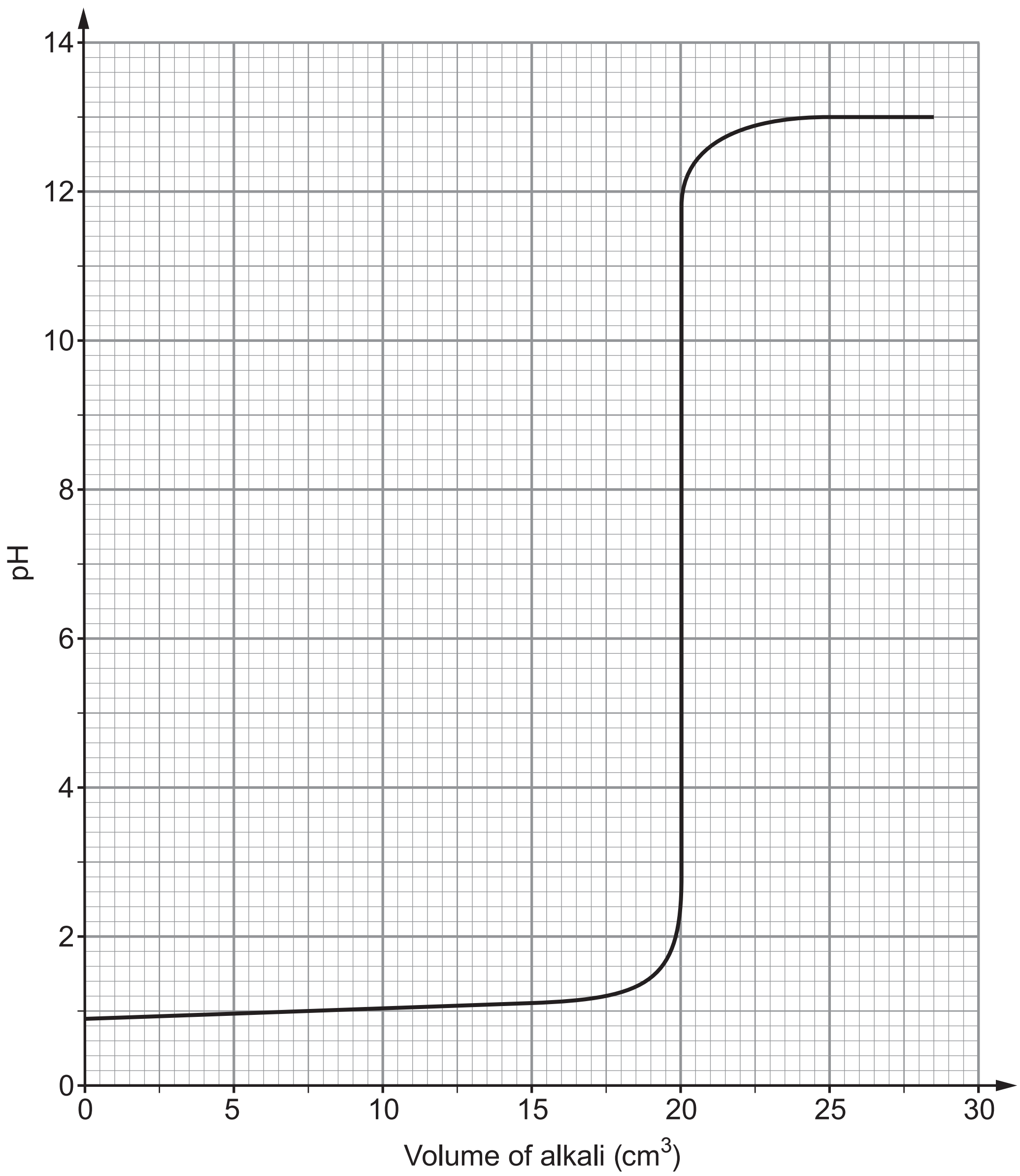
....

(ii) Calculate the mean titre. [1]

Mean titre = ... cm^3

Examiner only

(c) The graph below shows the pH changes, measured by a pH probe, that occurred during trial 4 of the experiment.



Explain the pH changes shown on the graph. [3]

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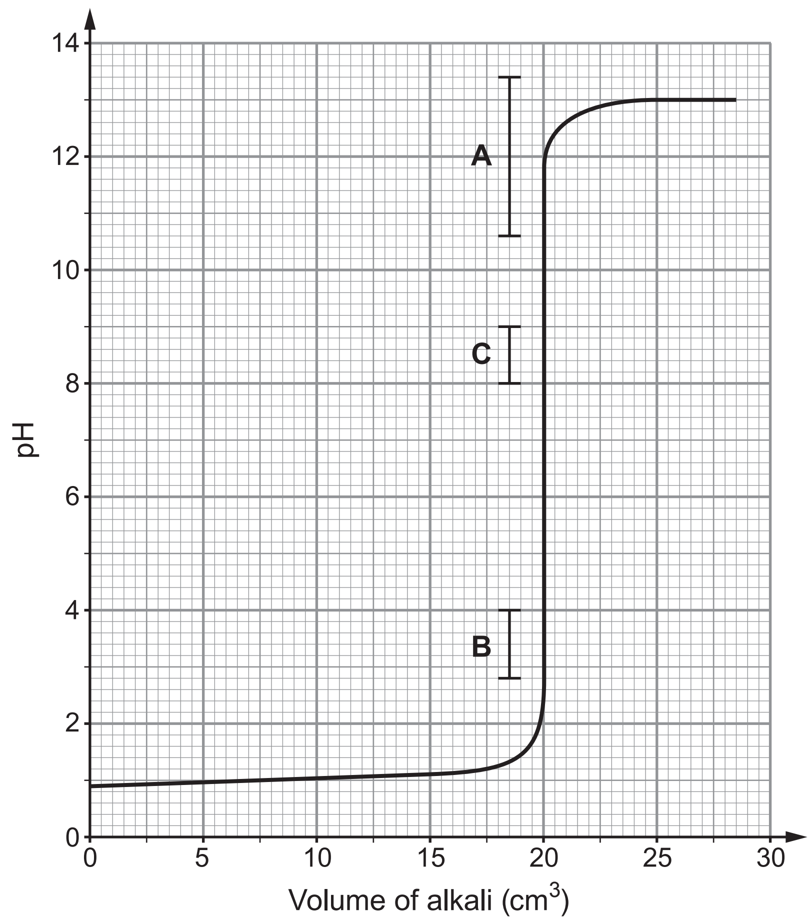
Examiner only

(d) Not all the students had a pH probe. Some had to use an indicator. The pH range of three different indicators is marked on the graph as **A**, **B**, and **C**.



(i) Use the information in the table below to match the range to each indicator. [1]

Indicator	pH range	Colour change	Range on the graph (**A**, **B**, or **C**?)
methyl blue	10.6 - 13.4	blue to violet	....
phenolphthalein	8.0 - 9.0	colourless to pink	....
dinitrophenol	2.8 - 4.0	colourless to yellow	....

(ii) Explain which indicator is not suitable for this experiment. [2]

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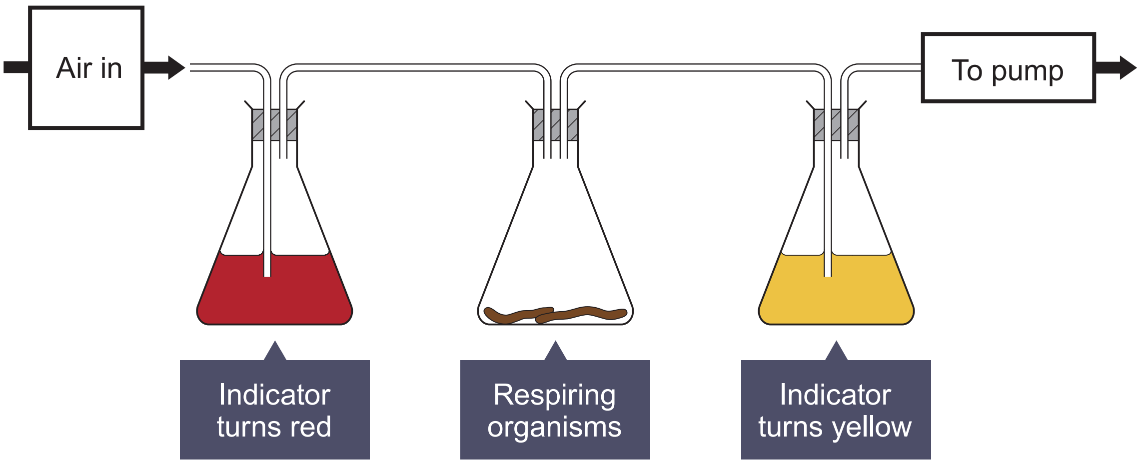
10

2. Hydrogen carbonate indicator can be used to measure the rate of respiration.

Concentration of carbon dioxide (%)	Colour of hydrogen carbonate indicator
less than 0.04	purple
0.04 (normal concentration in air)	red
greater than 0.04	yellow

The apparatus below can be used to compare the rate of respiration in small animals.



A group of students used different respiring organisms and recorded the time taken for the hydrogen carbonate indicator to turn yellow. The apparatus was kept at a constant temperature of 20 °C.
The results are shown in the table below.

Organism	Time taken for the hydrogen carbonate indicator to turn yellow (min)
earthworm	22
woodlouse	18
grasshopper	10
mouse	3

Examiner only

(a) Explain why the hydrogen carbonate indicator changed colour in this experiment. [2]

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(b) Give **two** reasons why the hydrogen carbonate indicator turned yellow fastest with the mouse. [2]

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(c) The experiment was performed at a temperature of 20 °C. [2]

State **two** other variables that need to be controlled when comparing the respiration rates of different organisms.

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(d) Describe how you would set up a control for this experiment and state the expected result. [3]

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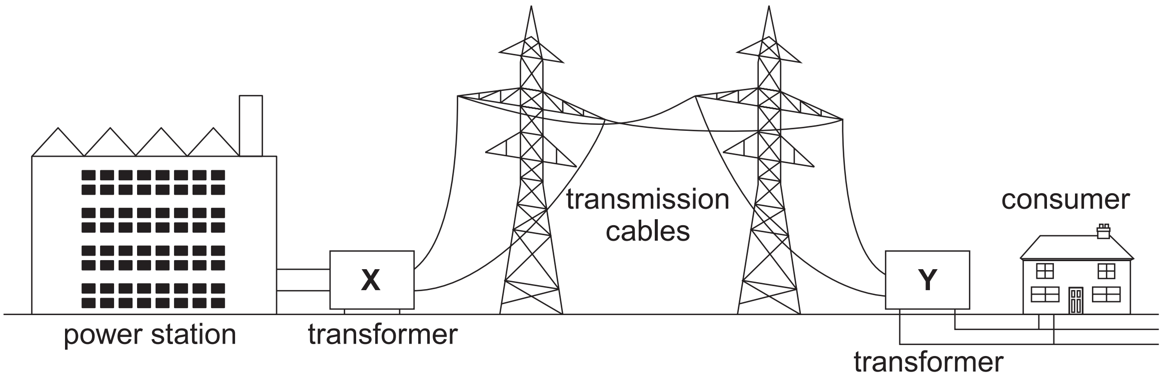
3445UA01 07

9

Examiner only

3. A diagram of part of the National Grid is shown below.



(a) Describe the role of the National Grid. [2]

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(b) Explain why transformers **X and Y** are used in the National Grid. [4]

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Examiner only

(c) Not all consumers can access the National Grid. An isolated farm with 5 sheds, in a rural part of mid-Wales is not connected to the National Grid.

The farmer needs to generate electricity for his sheds and he chooses a biomass generator that uses wood chips and pellets as an energy source and has a capacity of 100 kW.

Each of the sheds is supplied with 230 V with a maximum current of 80 A.

The farmer thinks that this generator will supply enough power for his 5 sheds.

(i) Use the equation:

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to determine if the 100 kW generator will be able to supply the 5 sheds. [4]

....

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(ii) Explain the environmental advantages of using the biomass generator. [3]

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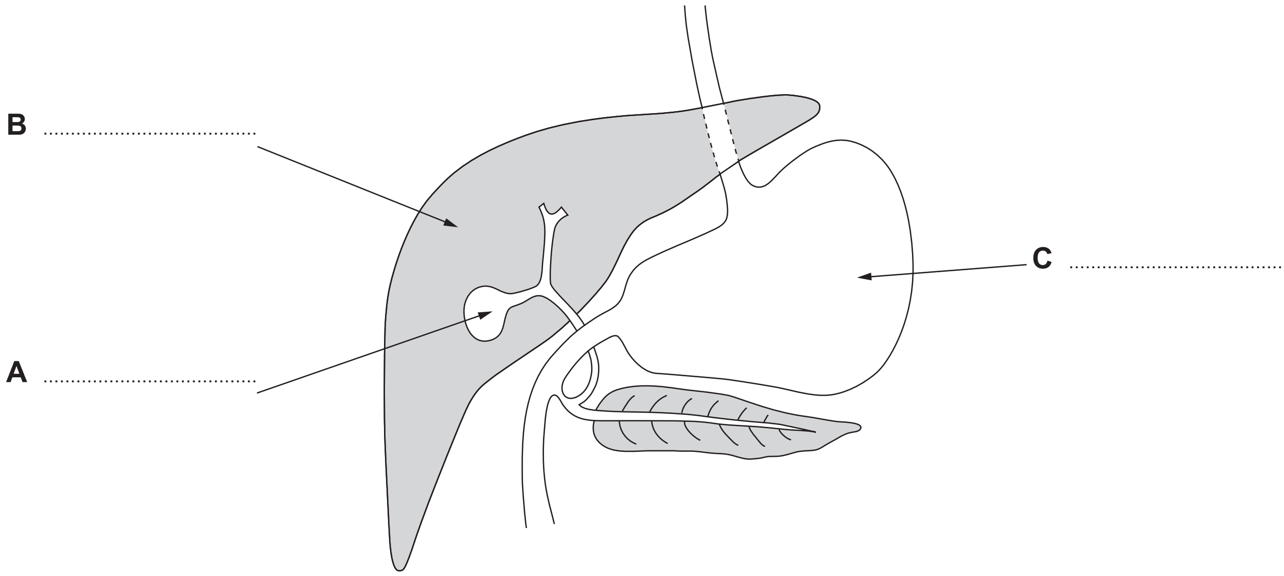
3445UA01 09

13

Examiner only

4. Food provides energy for the body. The digestive system is adapted to break down and absorb food. Parts of the digestive system are responsible for the digestion of foods by specific enzymes. The pH varies in these parts to aid the action of these enzymes in the digestion of the food.

(a) The diagram below shows part of the digestive system.



(i) Label parts **A**, **B** and **C**. [3]

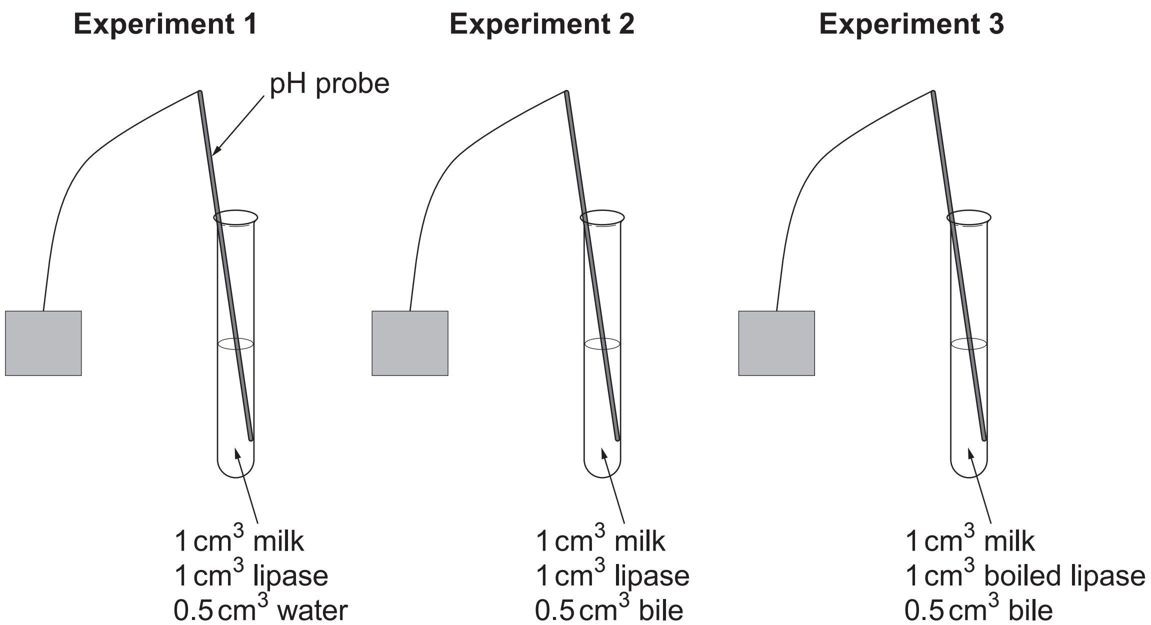
(ii) Name the organ that secretes the enzyme lipase. [1]

....

(b) A group of students was investigating the digestion of milk fat by the enzyme lipase. Lipase has an optimum pH of 10.

The students set up three experiments.

- In each experiment they mixed an alkaline solution of milk with lipase and either bile or water.
- They used the same volume of milk, lipase and bile or water in each experiment.
- They recorded the pH after mixing for 5, 10 and 20 minutes.



In each experiment the pH was recorded. The results of the experiments are shown in the table below.

Time (minutes)	pH		
	Experiment 1	Experiment 2	Experiment 3
0	10	10	10
5	10	8	10
10	9	7	10
20	8	6	10

Examiner only

(i) Explain the pH changes in experiments **1** and **2**. [4]

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(ii) Explain the pH recorded in experiment **3**. [2]

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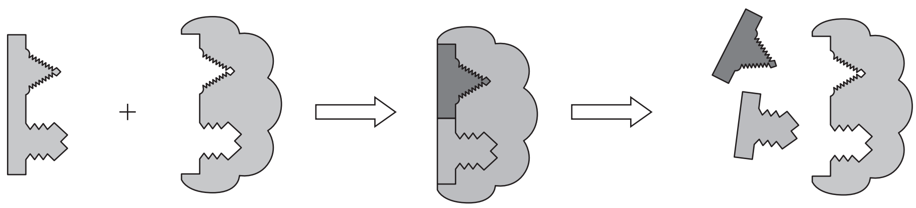
(c) The diagram below shows a model used to explain enzyme action.



Explain how this model works in relation to the digestion of lipids. [3]

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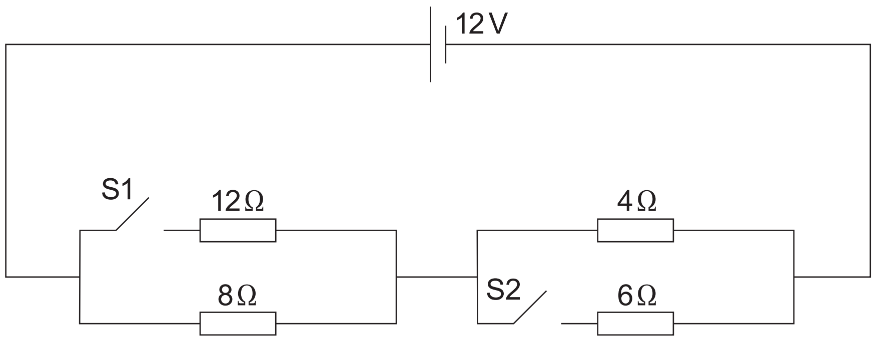
13

Examiner only

5. Students investigated the following electrical circuits.



Select the appropriate equations to answer the following questions.

voltage = current × resistance

$R = R_1 + R_2$

power = current^2 × resistance

$$\frac{1}{\text{total resistance}} = \frac{1}{\text{total resistance X}} + \frac{1}{\text{total resistance Y}}$$

(a) Calculate the current supplied when both switches are open. [3]

Current = ... A

(b) (i) Calculate the total resistance in the circuit. [3]

Total resistance = ... Ω

Examiner only

(ii) Calculate the power dissipated as heat by the 12 Ω resistor. [3]

Power = ... W

9

Examiner only

6. Water treatment is an important process in producing water for drinking. Temporary hard water contains calcium hydrogen carbonate $Ca(HCO_3)_2$ and magnesium hydrogen carbonate $Mg(HCO_3)_2$. Boiling breaks down the hydrogen carbonate to form insoluble carbonates. It is an inexpensive method used to soften temporary hard water.

(a) (i) Complete the equation below to show this process. [2]

$Ca(HCO_3)_2 \longrightarrow$ .... + ... + H_2O

(ii) Explain why permanent hard water cannot be softened by boiling. [2]

....

....

....

(b) The addition of sodium carbonate or passing water through an ion exchange column can be used to soften temporary and permanent hard water.

Explain how the following methods soften water.

(i) Addition of sodium carbonate. [2]

....

....

....

(ii) Passing water through an ion exchange column. [2]

....

....

....

(c) State **one** disadvantage other than cost associated with an ion exchange column. [1]

....

....

(d) Describe the effects of hard water on the central heating system in a house. [2]

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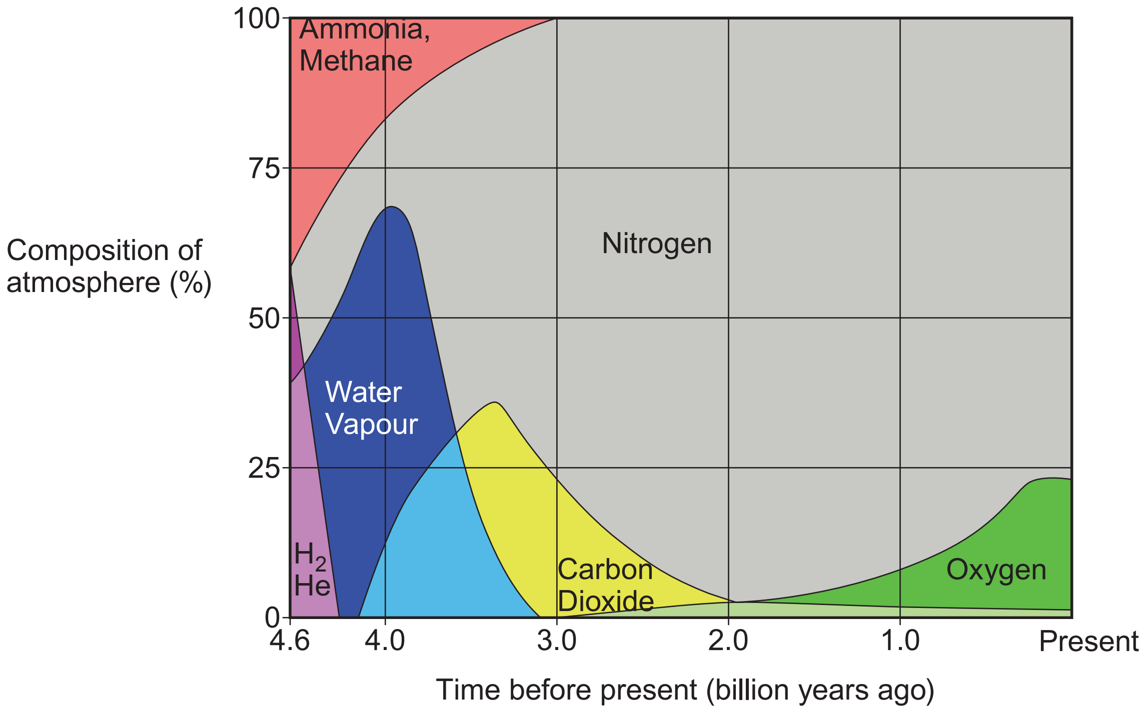
11

Examiner only

7. The graph below shows how the composition of the atmosphere has changed over time, since the formation of the Earth 4.6 billion years ago.



(a) Describe and explain how the composition of the atmosphere has changed over time. [6 QER]

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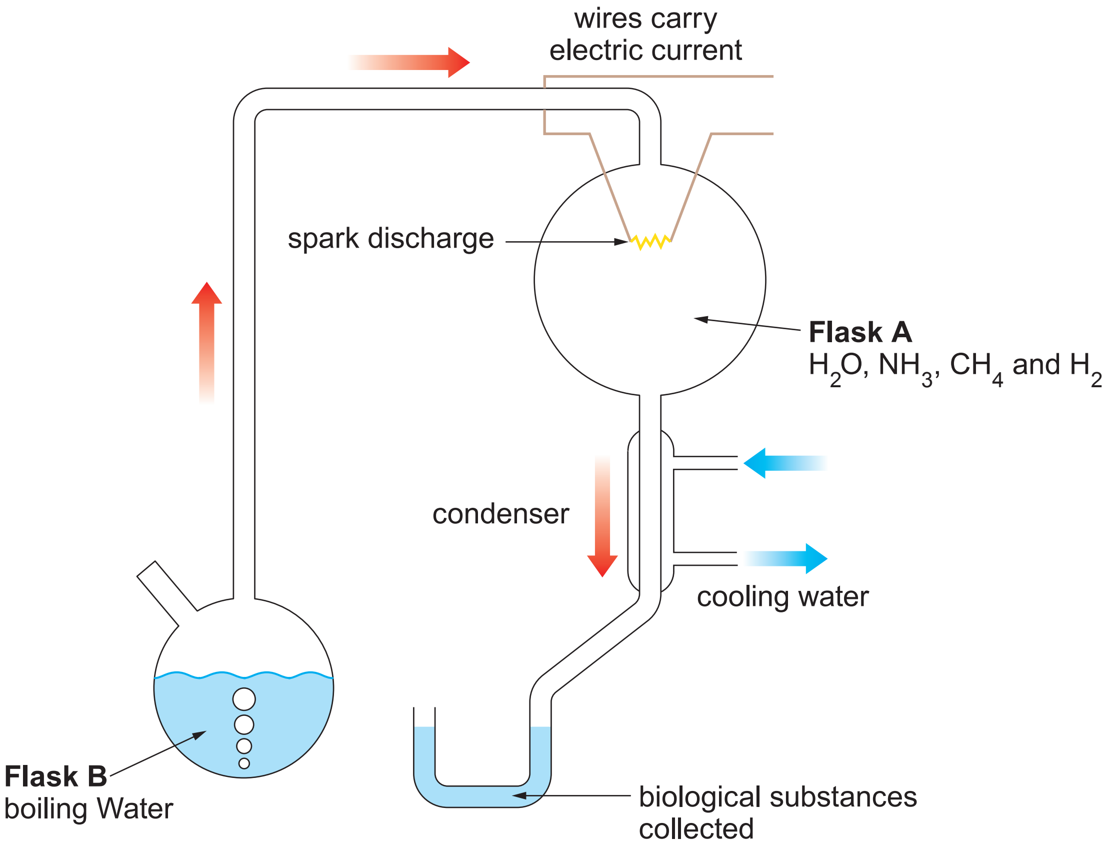
Examiner only

Examiner only

(b) In 1952 two scientists, Stanley Miller and Harold Urey, carried out experiments to see if the building blocks of common biological substances like carbohydrates (e.g. $C_6H_{12}O_6$), fatty acids (e.g. $CH_3(CH_2)_{17}COOH$), and amino acids (e.g. H_2NCH_2COOH) could be formed in the atmosphere of early Earth.

The scientists set up the apparatus shown below.



(i) Explain why Miller and Urey, chose the gases in **Flask A**. [2]

....

....

....

(ii) Use the information in the diagram to state **two** assumptions that Miller and Urey made about the early Earth other than the composition of the atmosphere. [2]

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10

END OF PAPER

THE PERIODIC TABLE

1	2					Group						3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number

Surname		Centre Number	Candidate Number
Other Names			0

GCSE

3445UA0-1

S19-3445UA0-1

APPLIED SCIENCE (Double Award)

UNIT 1: Energy, Resources and the Environment

HIGHER TIER

WEDNESDAY, 12 JUNE 2019 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**11**	
2.	**8**	
3.	**11**	
4.	**11**	
5.	**6**	
6.	**12**	
7.	**16**	
Total	**75**	

3445UA01
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 16.

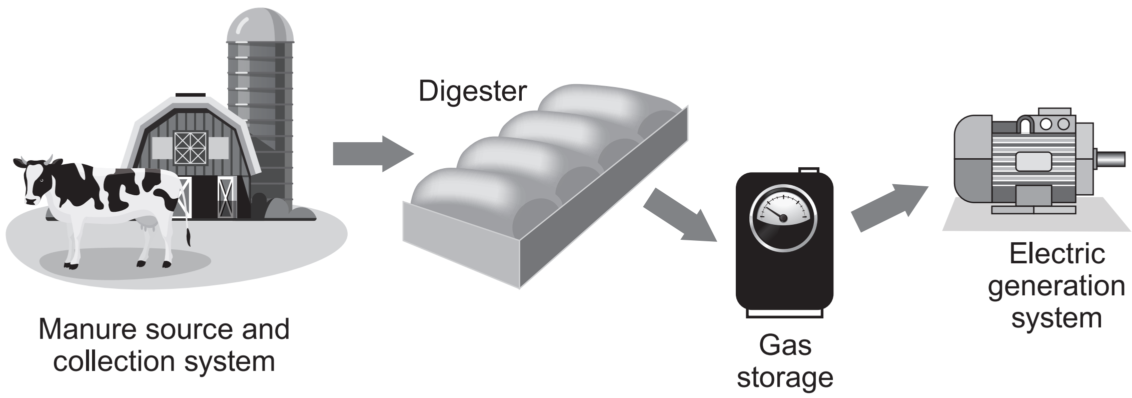
Examiner only

Answer **all** *questions.*

1. A farmer has installed a biogas generator to save money. The farmer's monthly electricity bill before installation was £3 000. The farmer spent £120 000 to buy and install the biogas generator. After installation the monthly electricity bill was expected to reduce to £600.

The biogas generator works by using the animal waste produced on the farm. This waste is digested by bacteria and the product, methane gas, is used to generate electricity. Methane is a greenhouse gas and produces carbon dioxide and water when it is burned

The animal waste used in the digester is a renewable energy source.



(a) The biogas generator can provide some of the farm's electricity.

Use the information above to calculate the expected payback time for the biogas generator. [3]

Payback time = ... months

(b) The farmer considered installing wind turbines to generate electricity before investing in the biogas generator.

Give **two** disadvantages of using wind power rather than the biogas generator. [2]

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Examiner only

(c) The biogas generator needs to produce 160 000 kWh of electricity per year to power the farm. A typical cow produces 10 tonnes of waste per year of which the farmer is only able to collect 3 tonnes. For every tonne of waste collected 480 kWh of heat energy is produced, which generates 160 kWh of electricity.

(i) Use the equation:

$$\% \text{ Efficiency} = \frac{\text{energy usefully transferred}}{\text{total energy supplied}} \times 100$$

to calculate the efficiency of this biogas generator [2]

% Efficiency = ...

(ii) The farmer owns 120 cows. He thinks that he will be able to collect enough waste to power his farm for a year. [4]

Explain whether you agree with the farmer. Show your workings as part of your answer.

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3445UA01
03

Examiner only

2. In a diffusion experiment, gelatin was mixed with a pink pH indicator and allowed to set. The gelatin was then cut into different sized cubes. The cubes were placed in a beaker containing dilute acid. As the acid diffused through the gelatin the cubes became colourless. The time taken to change from pink to colourless was recorded.

The surface area and volume of the cubes were calculated using the following equations:

surface area of cube = length × width × number of sides

volume of cube = length × width × height

(a) The results of the experiment are shown below.

Length of side of gelatin cube (cm)	Time taken to become colourless (s)	Surface area of cube (cm^2)	Volume of cube (cm^3)	Surface area:volume ratio
1	20	6	1	6:1
5	41	150	....	6:5
7	104	....	343	6:7
10	610	600	1 000	....

(i) **Complete** the table. [3]

(ii) Sarah suggested that diffusion of the acid into the gelatin is quicker in the smaller cubes because the time taken for them to become colourless is less. Explain whether you agree with Sarah. [2]

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Examiner only

(b) Red blood cells transport oxygen around the body to allow respiration to occur in the tissues. Red blood cells have a volume of 90 units and a surface area of 136 units. Explain how the surface area:volume ratio of red blood cells make them suitable for their function. [3]

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3445UA01 05

Examiner only

3. In 1912, Alfred Wegener suggested that all of the continents on Earth were once joined together in one supercontinent called Pangaea. These continents have since drifted apart. Wegener suggested a hypothesis called 'Continental Drift'. In the 1960's scientists further developed Wegener's hypothesis using a new theory called 'Plate Tectonics'.

(a) (i) State **three** pieces of evidence that Wegener used to support his hypothesis of *'Continental Drift'.* [3]

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(ii) Explain the theory of *'Plate Tectonics'*. [3]

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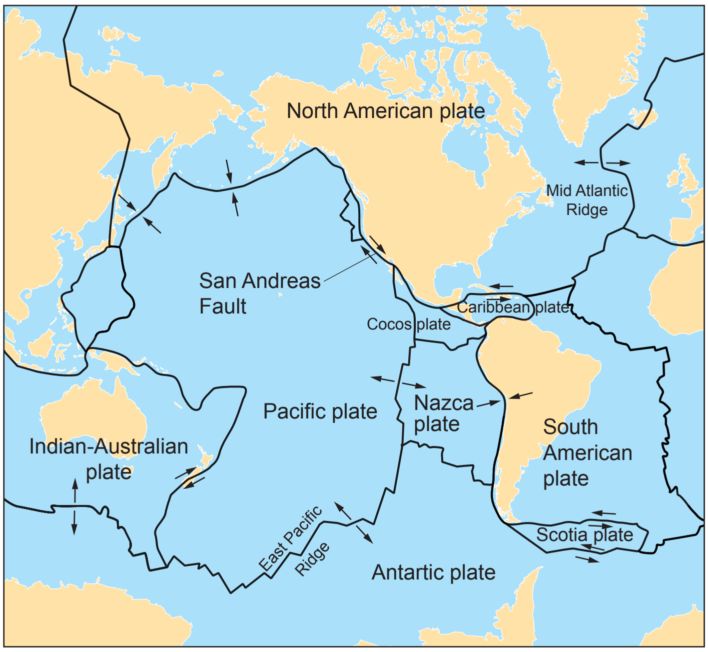
Examiner only

(b) The map below shows some of the tectonic plates on Earth and the arrows show the direction of their movement.



Tectonic plates move relative to each other and are monitored closely by geologists using a seismometer. The Pacific and North American plate are very closely monitored by the US government as large populations of people can be affected, especially in the state of California. The boundary between the Pacific and North American tectonic plates is described as a conservative boundary where the plates slide past each other causing powerful earthquakes. The San Andreas Fault lies on this boundary.

Tectonic plate boundaries can also be described as constructive and destructive.

(i) **Label** on the map **one** constructive (C) and **one** destructive (D) tectonic plate boundary. [1]

(ii) Describe these **two** types of boundary and the **consequences** of their movement.

Destructive .... [2]

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Constructive .... [2]

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3445UA01 07

11

Examiner only

4. Port Talbot steel works in South Wales is one of the largest in the UK. It produces 5 million tonnes of steel every year.

The materials required for the production of one tonne of iron are shown in the table below. The melting point of iron is 1535 °C.

(a) Describe the purpose of the following raw materials in the blast furnace. [2]

Haematite ....

Limestone ....

(b) The blast furnace relies on the processes of reduction and oxidation. In one stage carbon monoxide is formed from coke.

(i) **Complete** the balanced symbol equation for the extraction of iron in the blast furnace. [3]

Fe_2O_3 + ... $\longrightarrow$ .... + ...

(ii) State the substances that are reduced or oxidised in the reactions that must take place to produce iron in the blast furnace. [2]

....

....

....

(c) The materials required for the production of one tonne of iron are shown in the table below. The melting point of iron is 1535 °C.

Raw material	Mass of raw material (tonnes)	Cost per tonne of raw material (£)
haematite (iron ore)	1.75	60.50
coke	0.25	120.90
limestone	0.25	80.00
air	4.00	2.25

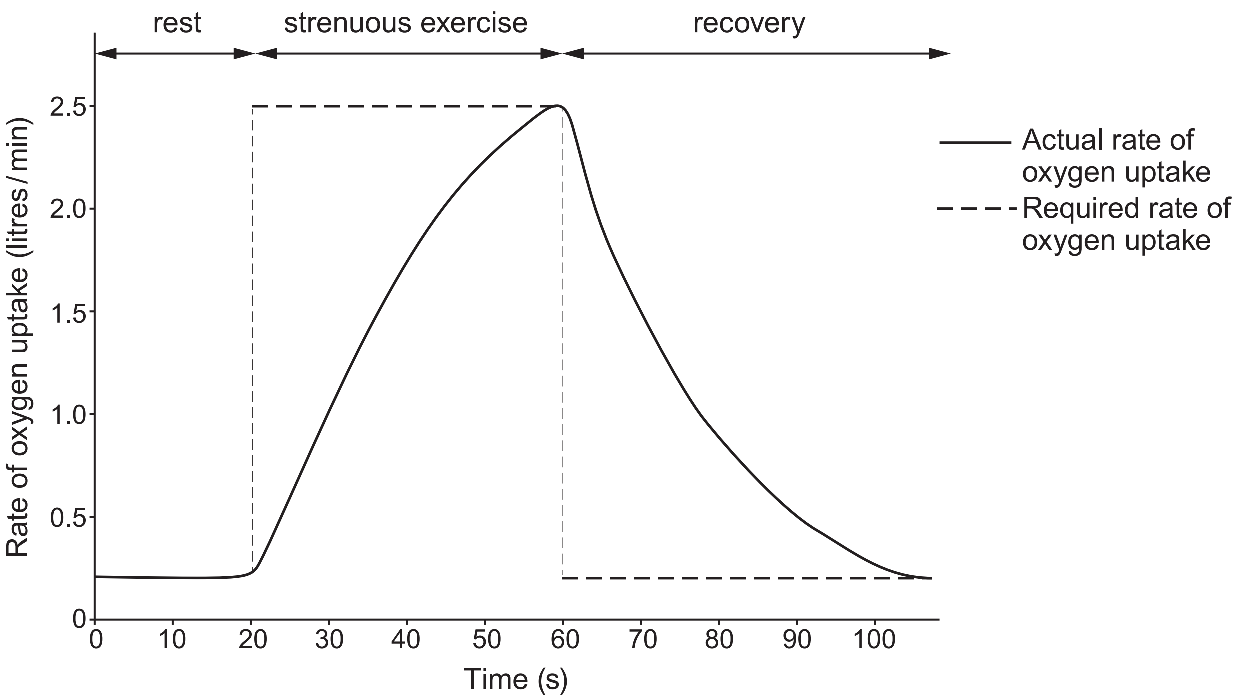
Calculate the **monthly** raw material cost, for the Port Talbot steel works. Give your answer to two significant figures. [4]

= £ .... million / month

11

Examiner only

5. Respiratory technicians are researching the effect of strenuous exercise on an athlete. They measure how the rate of oxygen uptake can change in an athlete. They do this before, during and after strenuous exercise. This is shown in the graph below.



Use your knowledge and the information in the graph to explain how the type of respiration changes during the 100 seconds shown. [6 QER]

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6

Examiner only

6. Digestion of food is required to provide nutrition for the human body.

(a) (i) Explain how digestion provides nutrition. [2]

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....

(ii) Describe the process of peristalsis in the digestive system. [2]

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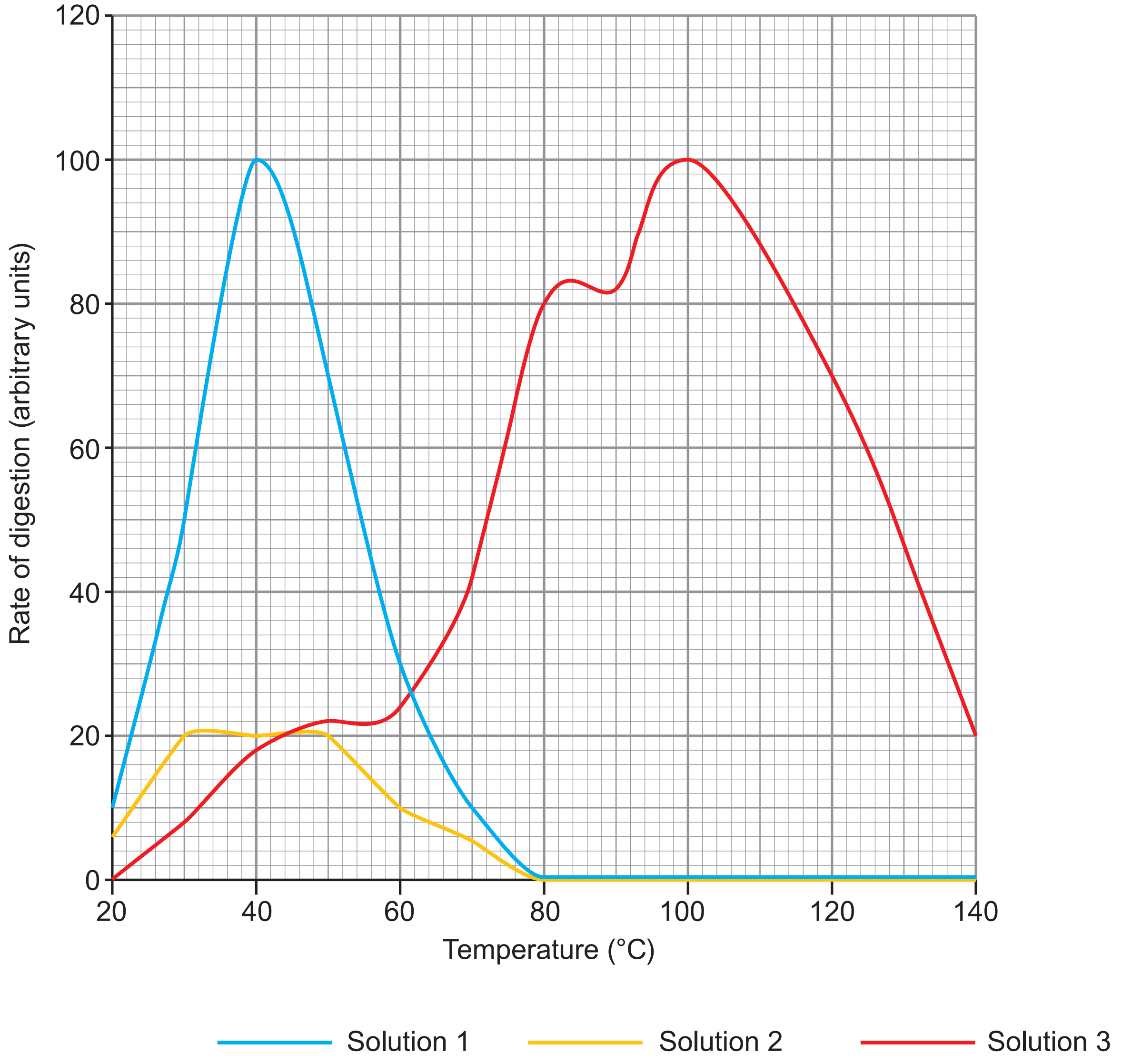
....

(b) A group of students were given two solutions of a human enzyme **(solution 1 and solution 2)** and one solution of bacterial enzyme **(solution 3)**. All solutions were the same concentration and volume. They measured the rate of digestion using each enzyme solution. A computer calculated the rate of digestion and the results are shown below.



Examiner only

Use the data in the graph and your knowledge to answer the following questions.

(i) Using the results for **solution 1**, describe and explain the effect of temperature on the rate of digestion between 20 and 60 °C. [4]

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(ii) Suggest **one** reason why the results shown for **solution 2** are different to **solution 1**. [1]

....

....

(iii) State a suitable control in this experiment. [1]

....

(iv) **Solution 3** contains an enzyme from a species of bacteria that is found in a hot volcanic region.

Describe the differences between the action of this bacterial enzyme and the enzyme found in humans. [2]

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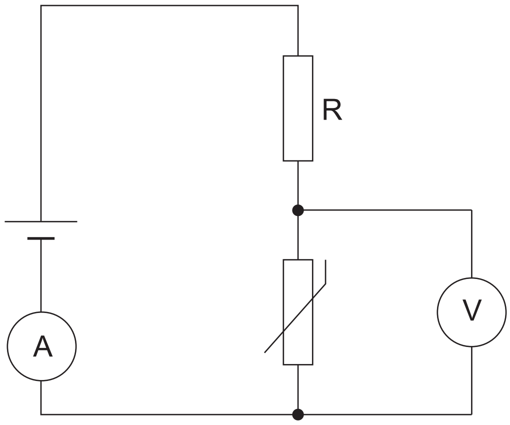
12

Examiner only

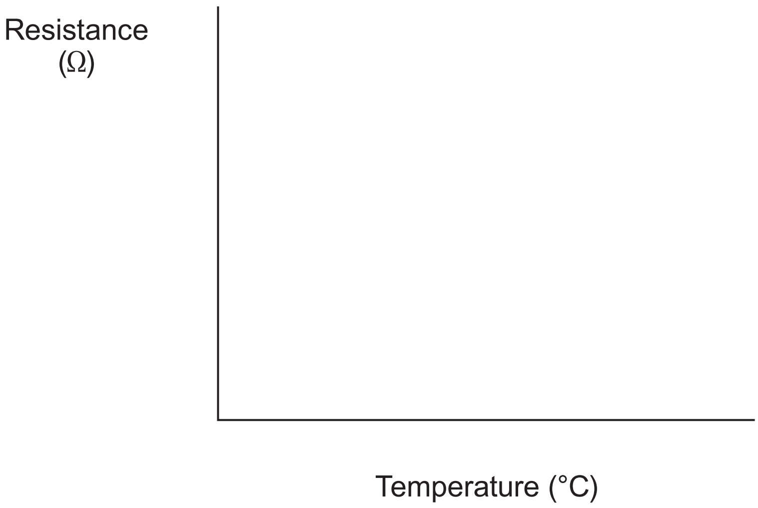
7. Students were investigating the properties of circuits containing LDRs and thermistors.

(a) The students set up the following circuit.



The temperature of the thermistor was increased.

(i) On the axes below, sketch how you would expect the resistance of the thermistor to change as the temperature increased. [1]



Examiner only

(ii) Explain how increasing the temperature of the thermistor affects the readings on the ammeter and voltmeter. [5]

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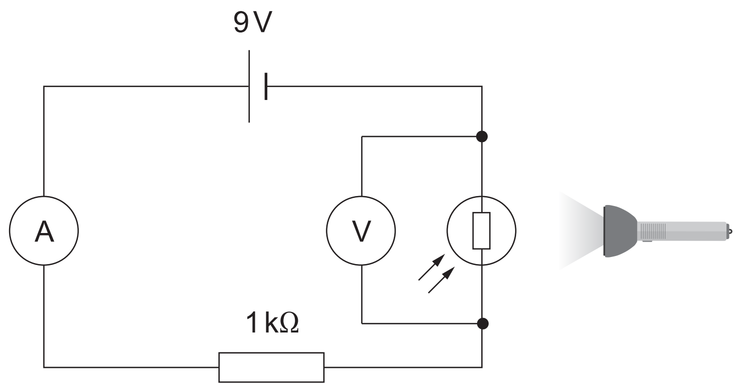
Examiner only

(b) The students then investigated an LDR with a torch and the following circuit.



(i) State the independent variable in this experiment. [1]

....

(ii) State the dependent variable in this experiment. [1]

....

(iii) Use the equations:

$$R = R_1 + R_2$$

and

$$V = IR$$

to calculate the readings on the ammeter and the voltmeter in the circuit when the resistance of the LDR is 2.5 kΩ. [5]

Ammeter: .... A

Voltmeter: V

Examiner only

(iv) Describe how you would modify this investigation to determine the I-V characteristics of the LDR. [3]

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16

END OF PAPER

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3445UA0-1

Z22-3445UA0-1

MONDAY, 20 JUNE 2022 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

HIGHER TIER

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**19**	
2.	**8**	
3.	**6**	
4.	**14**	
5.	**12**	
6.	**16**	
Total	**75**	

3445UA01
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

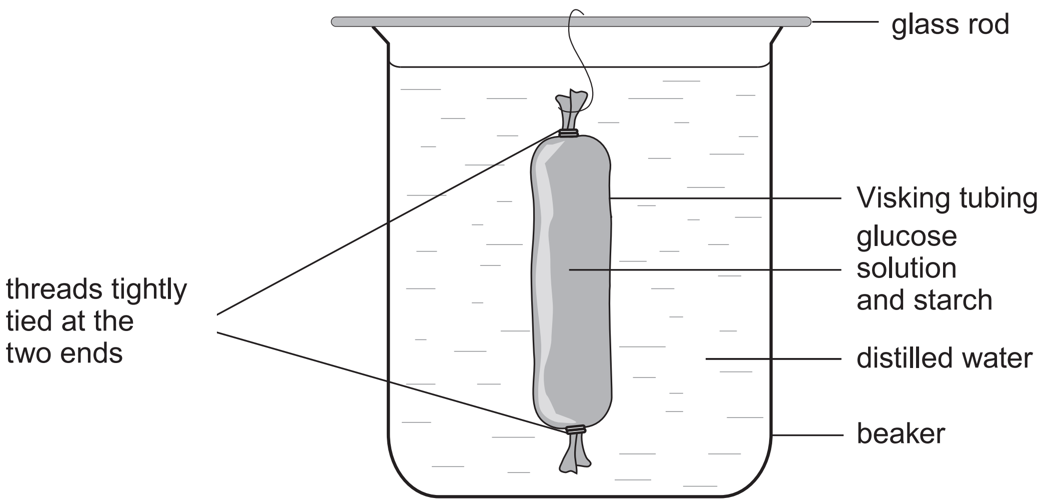
The number of marks is given in brackets at the end of each question or part-question.
Question **3** is a quality of extended response (QER) question where your writing skills will be assessed.
You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
A Periodic Table is printed on page 20.

JUN223445UA0101

Examiner only

Answer **all** questions.

1. A student set up the following experiment to show how diffusion of digested foods occurs in the small intestine. The Visking tubing contained starch and a solution of glucose with a concentration of 100 g/dm^3. This was placed in a beaker containing distilled water at 20 °C and left for 2 hours.



(a) (i) Explain what is expected to happen to the starch and the glucose in the Visking tubing during the experiment. [4]

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(ii) Describe **two** chemical tests the student should carry out on the contents of the beaker to confirm your answer to (a)(i) and state the expected results. [4]

1. ....

....

....

2. ....

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Examiner only

(b) Explain how each of the following changes would affect the diffusion process in this experiment.

(i) Carbohydrase is added to the contents of the Visking tubing. [2]

(ii) The investigation is carried out at 35 °C. [2]

(iii) The experiment is carried out with a solution of glucose of 100 g/dm^3 concentration replacing the distilled water in the beaker. [2]

3445UA01 03

(c) State **one** of the limitations of using Visking tubing as a model intestine. [1]

(d) Explain **two** adaptations of the small intestine for the absorption of digested food. [4]

1.

2.

19

Examiner only

2. Sodium oxide is formed when sodium burns in oxygen.

(a) Use the Periodic Table on page 20 to describe the structure of the oxygen atom. [3]

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(b) During the production of sodium oxide the Na^+ and O^{2-} ions are formed.

(i) Give the electronic structures of the ions. [2]

Na^+ ....

O^{2-} ....

(ii) Sodium reacts violently with water to form a strong alkali.
State the name of the alkali formed in this reaction. [1]

....

(c) Small amounts of sodium peroxide, Na_2O_2, are also formed when sodium burns.
Calculate the relative formula mass (M_r) of sodium peroxide. [2]

A_r: Na = 23, A_r: O = 16

M_r = ...

8

Examiner only

3. Describe how the elements are arranged in the Periodic Table and explain, with examples, how their properties relate to their position. [6 QER]

6

3445UA01 05

Examiner only

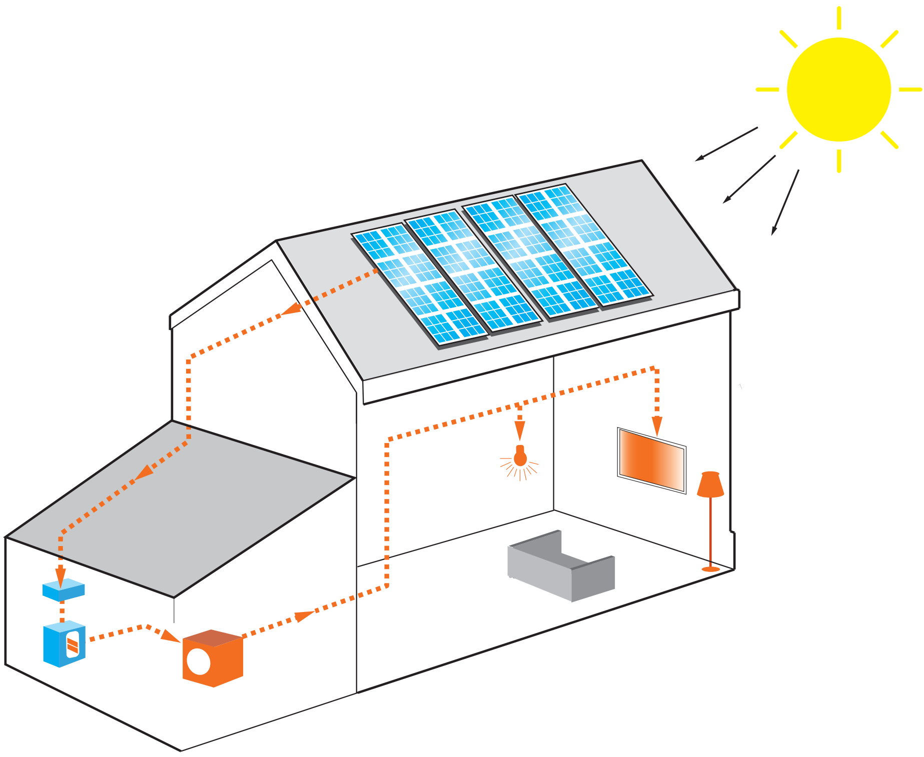
4. In the UK there are now a range of companies that will offer to install solar panels to homes.

Adverts like the one below are becoming quite common.

Sunny Sun Panels

You can save an average of £50 each month on your fuel bill!

When the Sun's radiation hits your solar panels it is converted into electrical energy. This energy travels to a device called an inverter, which converts it into usable household electricity. With over 1500 hours of sunshine per year what could be better!



Your own Sunny Sun panel system fully installed for £3800! What a price!
Add battery storage at only £2900 – you can use your solar electricity whenever you like.

Get your money back in 10 years!

Designed to work alongside your existing electricity supplier or as part of a new combined home solar system, battery storage will make it much easier to save on your electricity bills by utilising more of the electricity generated by your solar panels. This system will satisfy all your home electrical needs.

The excess power from your solar panels charges your battery during the day. The battery then supplies electricity to your home during the night and on cloudy days. This maximises your electricity savings and makes your house free from electricity from the National Grid.

(a) In January 2018 a householder invested in the Sunny Sun panel system and battery storage.

(i) Use the equation

$$\text{payback time} = \frac{\text{installation costs}}{\text{annual savings}}$$

and the information provided to determine whether the claim 'get your money back in 10 years' is true. [3]

....

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(ii) State **three** reasons why payback time might be more than expected. [3]

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(b) In 2018 the mean household electricity consumption for the UK was 4420 kWh per year.

An independent energy expert assessed the system and found they absorb a mean power of 14 000 W of sunlight and converted this to electrical power with an efficiency of 23 %.

Sunny Sun claim that their panel system can satisfy the electrical needs of the average home in the UK.

Use this information together with information in the advert and the equation

$$\text{units produced (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to determine if the Sunny Sun claim is valid. [4]

....

....

Examiner only

(c) One of the largest solar power plants proposed in the UK is being developed in Kent. This solar power plant is capable of generating a maximum of 3.5×10^6 kWh per year. At present the UK generates 5.3×10^{10} kWh of electricity per year.

Use the information provided to:

(i) Calculate how many homes this solar power plant could supply with electricity assuming that the mean household electricity consumption is 4420 kWh per year. [2]

number of homes = ...

(ii) Calculate how many solar power plants of this size would be required to supply the UK electrical energy demand. [2]

number of plants = ...

14

Examiner only

5. Pancreatitis is a disease that causes inflammation of the pancreas and can result in a very low rate of enzyme production and secretion into the small intestine. Enzymes can be given to patients as a treatment for this disease. The enzymes are taken as a slow-release capsule which is not affected by acid.

(a) State the function of an enzyme. [1]

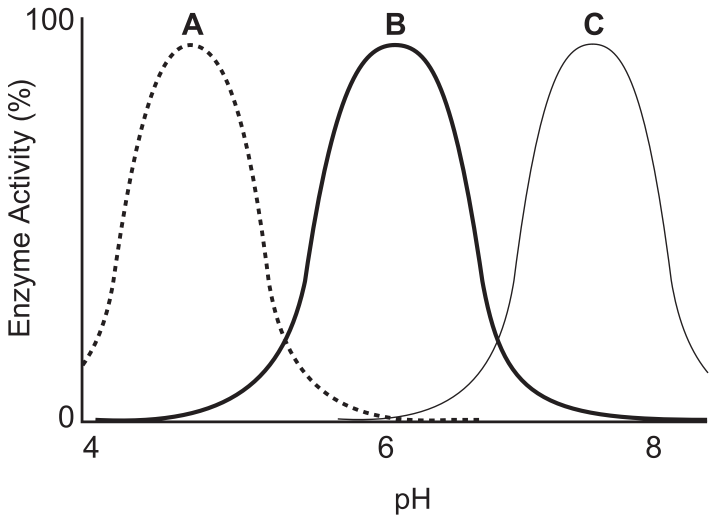
....

(b) Suggest a reason why the enzyme is not given to the patient as a liquid. [2]

....

....

(c) The activity of enzymes **A**, **B**, and **C** at different pH levels are shown below.



Explain whether enzymes **A**, **B**, or **C** are suitable for use in treating a patient with pancreatitis. [4]

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Examiner only

(d) Enzymes are also used in the manufacture of biological washing powders. Biological washing powders contain proteases, carbohydrases and lipases.

(i) Suggest why washing powders need these three types of enzymes to remove food stains from clothes. [1]

....

....

(ii) Explain why washing at 80 °C using biological washing powders would not be recommended for removing food stains. [4]

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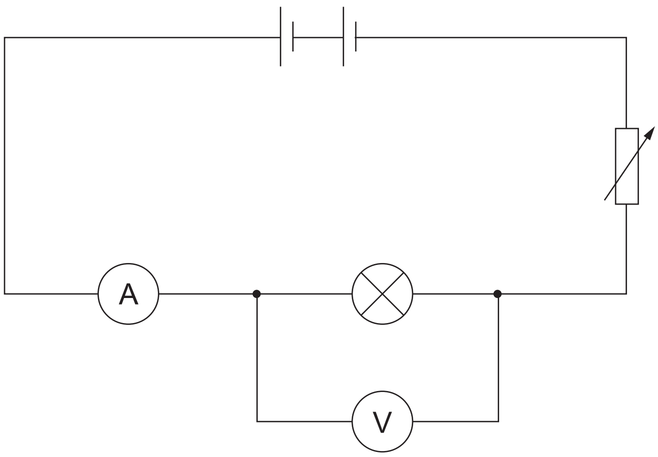
12

Examiner only

6. A student was investigating how current changes with voltage for various components.

She used the following circuit to investigate a lamp.



She followed the method below.

1. Connect the circuit as shown in the diagram.
2. Set the variable resistor to give the lowest voltage and record the readings on the voltmeter and ammeter.
3. Alter the variable resistor to increase the voltage by 0.5 V.
4. Record the new readings on the voltmeter and ammeter.
5. Repeat steps 3 and 4.

(a) Some of the results are shown below.

Voltage (V)	Current (A)
0.0	0.00
2.5	0.65
3.5	0.90
4.5	1.16
5.0	1.20
5.5	1.26
6.5	1.36
8.0	1.46

(i) State the independent variable in this experiment. [1]

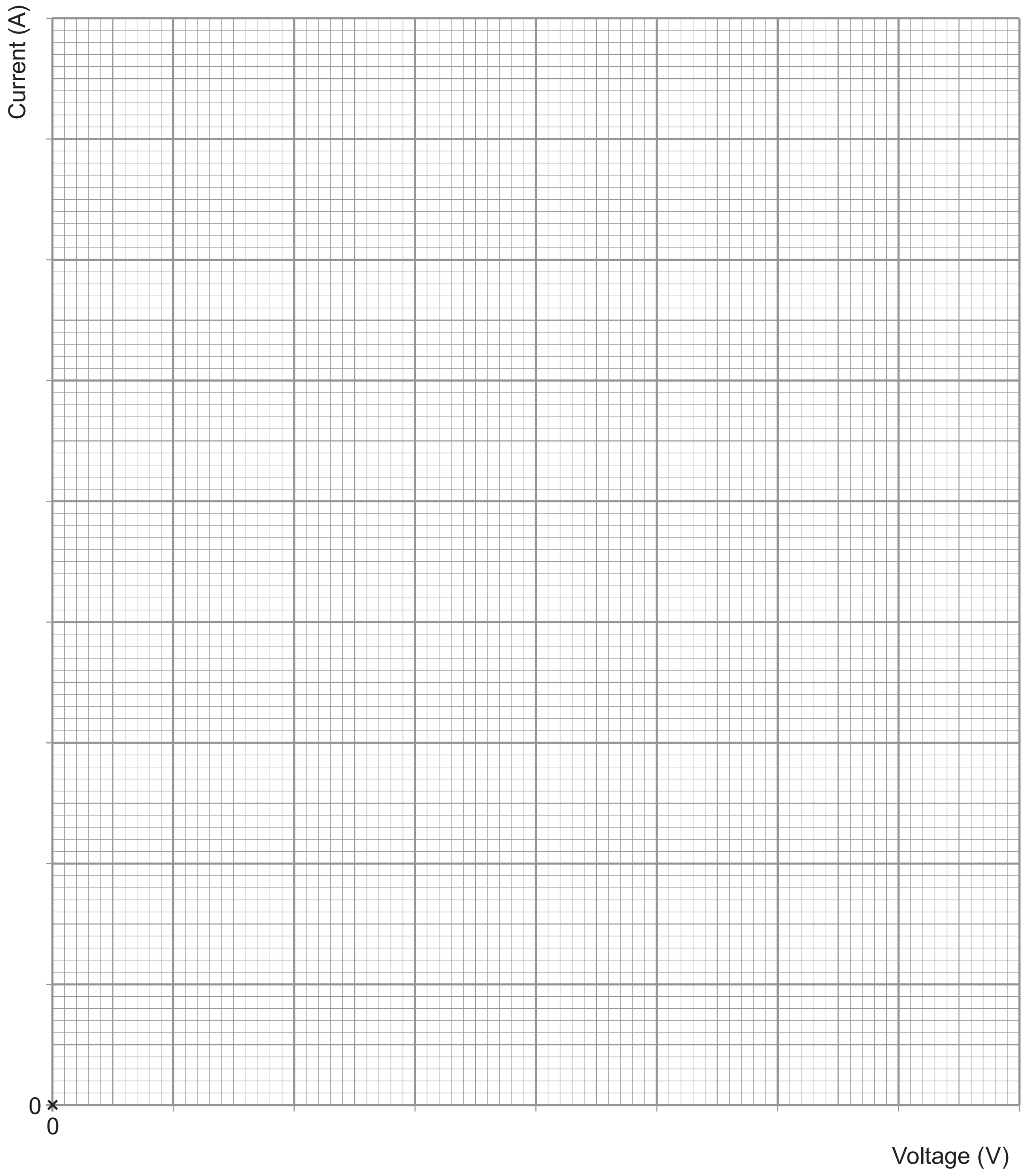
....

(ii) State the dependent variable in this experiment. [1]

....

Examiner only

(iii) Plot the results on the grid below and draw a suitable line. [4]



Examiner only

(iv) Use your graph in (a)(iii) and the following equation

voltage = current × resistance

to calculate the resistance of the lamp when the voltage is 7.2 V. [3]

resistance = ... Ω

(v) Explain whether the resistance of the lamp is constant throughout the experiment. [3]

....

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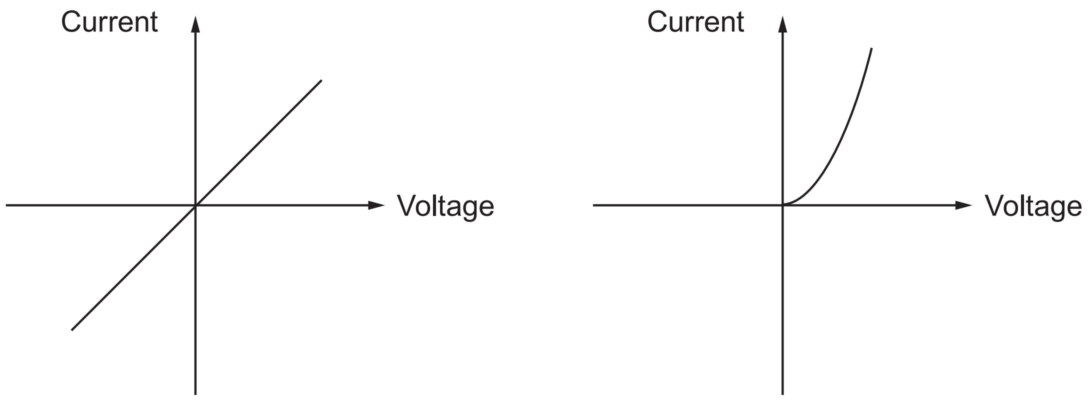
Examiner only

(b) The investigation was repeated with a resistor and then a diode.
The graphs of the results are shown below.



Use the graphs to compare the variation in resistance for these components. [4]

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16

END OF PAPER

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3445UA0-1

S23-3445UA0-1

FRIDAY, 16 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

HIGHER TIER

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**8**	
2.	**11**	
3.	**16**	
4.	**6**	
5.	**8**	
6.	**13**	
7.	**13**	
Total	**75**	

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01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **4** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A Periodic Table is printed on page 20.

Examiner only

Answer **all** questions.

1. Sioned reacted sulfuric acid with zinc oxide (ZnO).
She followed the method below to form zinc sulfate.

1. Pour 50 cm^3 of 1.0 mol / dm^3 sulfuric acid into a conical flask.
2. Heat the sulfuric acid to 50°C.
3. Add 5 g of zinc oxide to the warmed acid and stir.
4. Repeat step 3 until the zinc oxide is in excess.

(a) The main hazard in this experiment is that 1.0 mol / dm^3 sulfuric acid is an irritant.

(i) Describe **one** risk associated with using sulfuric acid. [1]

....

....

(ii) State a suitable control measure Sioned should carry out to reduce this risk. [1]

....

....

(b) Circle the correct formula in each bracket to complete the symbol equation for this reaction. [2]

$$ZnO + H_2SO_4 \longrightarrow \mathbf{(2ZnO_2\ /\ ZnSO_4\ /\ Zn_4S_2O)} + \mathbf{(O_2\ /\ H_2\ /\ H_2O)}$$

(c) State **two** additional steps that Sioned needs to carry out to obtain pure zinc sulfate crystals. [2]

1. ....

....

2. ....

....

Examiner only

(d) Robert has calculated the relative formula mass of zinc oxide to be 97.
Use the Periodic Table on page 20 to explain if he is correct. [2]

....

8

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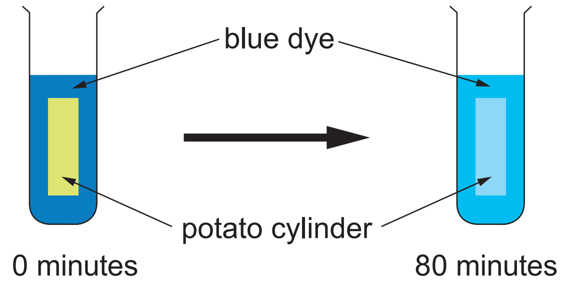
03

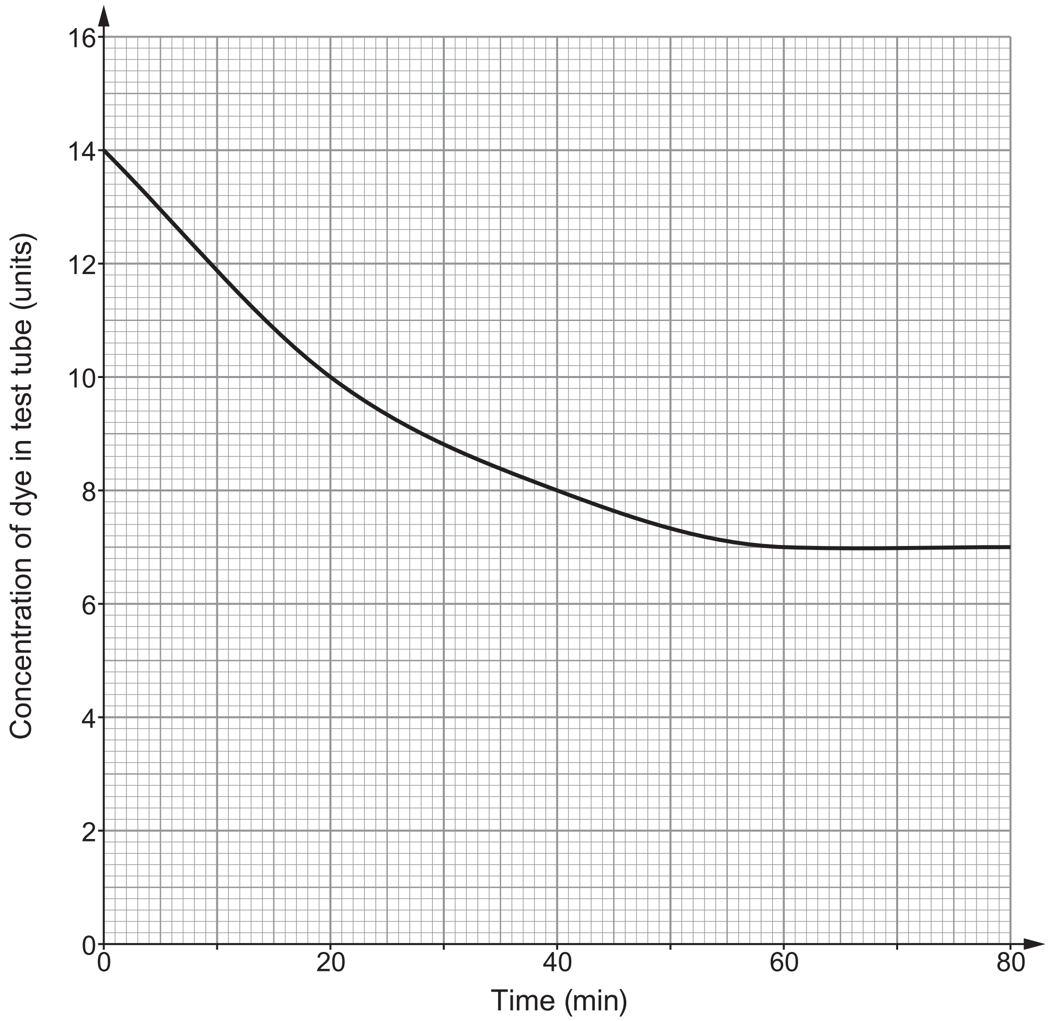
Examiner only

2. Several groups of students investigated the rate of diffusion of blue dye into potato by the method below.

1. Cut a cylinder of potato.
2. Place the cylinder in a test tube containing a solution of blue dye.
3. Put the test tube in a water bath at 20°C.
4. Measure the concentration of the dye after 20 minutes.
5. Continue to measure the concentration every 20 minutes up to 80 minutes.



The concentration of the dye left in the test tube after each 20-minute interval is shown on the graph below.



Examiner only

(a) (i) State the dependent variable in this experiment. [1]

....

(ii) State **two** variables that should be controlled. [2]

1. ....

2. ....

(b) (i) State how the concentration of dye left in the test tube changed during the first 20 minutes. [1]

....

(ii) Explain why this change occurred. [2]

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....

(c) One student suggests that the rate of diffusion is constant over the whole 80 minutes. Explain whether you agree. [3]

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(d) The students repeated the experiment in a water bath at 30°C. **Add another line** to the graph to show the expected results. [2]

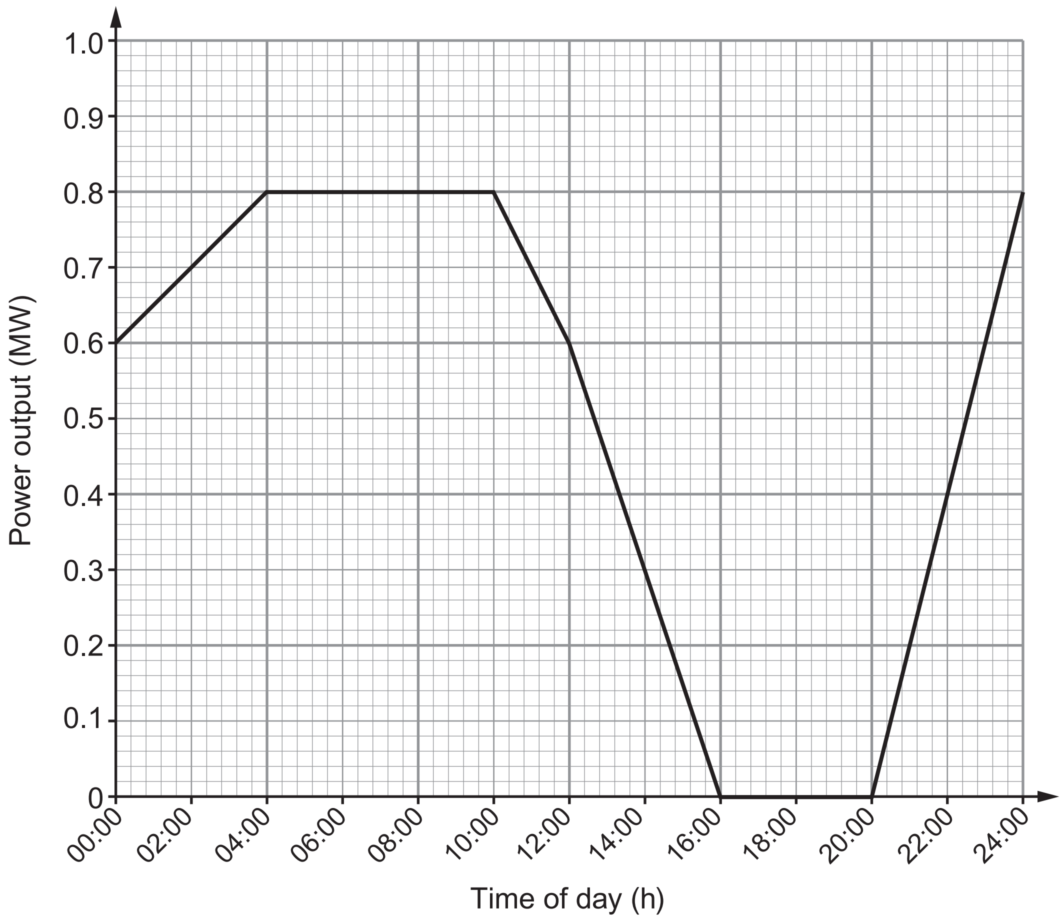
11

3445UA01 05

Examiner only

3. One renewable energy source that large energy suppliers use to generate electricity is wind. The percentage of electricity generated in this way has been gradually increasing over the last few years.

The graph below shows the power output of a wind turbine in one day. This wind turbine has a mean output of 0.6 MW.



(a) Use the graph and the equation:

units generated (kWh) = power (kW) × time (h)

to calculate the number of units generated between 04:00 and 10:00. [4]

(1 MW = 1000 kW)

number of units = ... kWh

(b) The table below gives information about generating electricity by wind and a nuclear power station.

	wind turbine	nuclear power station
expected lifetime (years)	20	60
mean power output (MW)	0.60	3 600
land area needed (km^2)	0.55	4.5
cost to commission (£ million)	3	4 000
waste produced	none	radioactive waste
lifetime carbon footprint (gCO2 / kWh)	5	5
overall cost of generating electricity (p / kWh)	5.6	2.8

Use the table to answer the following questions.

(i) Calculate the number of wind turbines that need to be built to provide the same power as the nuclear power station will provide in its lifetime. [2]

number of wind turbines = ...

(ii) Calculate the land area needed by a wind farm to produce the same power as one nuclear power station. [3]

land area = ... km^2

Examiner only

(c) (i) Evaluate the cost effectiveness of the two methods of generating electricity. [3]

(ii) Evaluate the environmental impact of the two methods of generating electricity. [2]

(d) The lifetime carbon footprints of a wind farm and a nuclear power station are 5 gCO2 / kWh.

Suggest how the carbon footprint of a coal-fired power station might compare to this value.
Give a reason for your answer. [2]

16

Examiner only

4. The diagrams below show the side view of the chest cavity, and a bell jar model that can be used to illustrate inspiration.

Diagram 1: Side view of the chest cavity

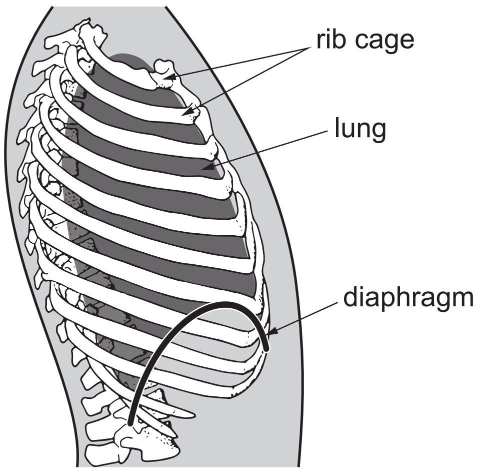
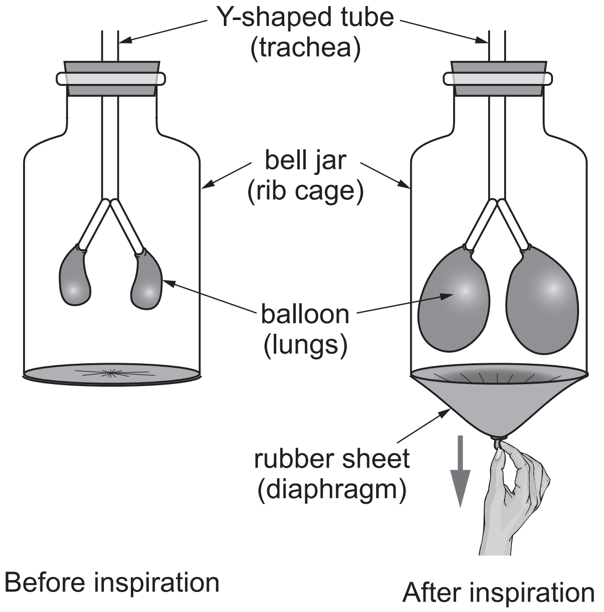


Diagram 2: Bell jar model



Describe the process of inspiration in the human body **and** explain the limitations of the bell jar as a model of this process. [6 QER]

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09

Examiner only

6

Examiner only

5. Developments in plastic technology have allowed the production of bioplastics using plants as the source of raw material.

The table below compares the production of the same mass of a bioplastic with polystyrene. Polystyrene is made from crude oil.

	Bioplastic	Polystyrene
Energy consumption (kWh)	10.7	22.6
Water used in production (dm^3)	74.6	185
Waste for landfill produced (g)	1.74	2.10
Biodegradable	Yes	No
Breakdown products	carbon dioxide, water and methane	styrene
kgCO2eq / kg produced	0.50	2.2

Use your knowledge and the information in the table to answer the following questions.

(a) Explain **two** advantages to the environment of using bioplastics. [4]

....

....

....

....

....

(b) Explain **one** environmental disadvantage of using bioplastics. [2]

....

....

....

(c) Manufacturers of this bioplastic claim that it is a carbon-neutral product. State the additional information that is needed to support this claim. [2]

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....

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8

Examiner only

6. Jamie is investigating series and parallel electrical circuits.

The following equations apply to electrical circuits:

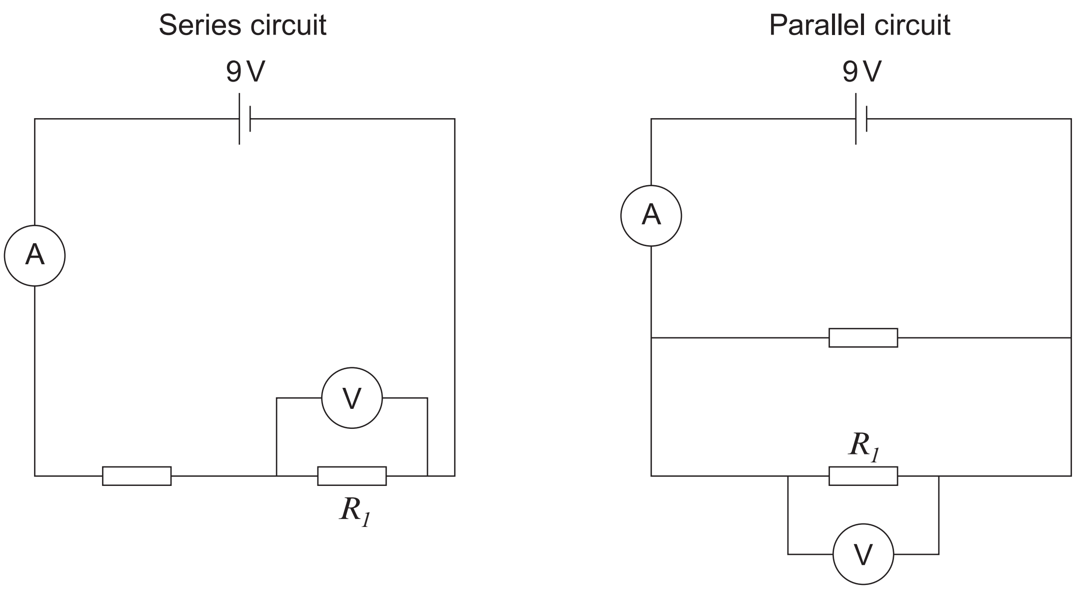
$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

$$\text{power} = \text{voltage} \times \text{current}$$

$$\text{power} = \text{current}^2 \times \text{resistance}$$

$$R_T = R_1 + R_2 \;\ldots\ldots\ldots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \;\ldots\ldots\ldots$$



(a) Jamie built the circuits above with identical resistors. In the series circuit the total resistance in 4.5 ohms and the ammeter reading is 2 Amps.

(i) I. Compare the total resistance in the parallel circuit with the series circuit. [2]

....

....

Examiner only

II. Compare the ammeter reading in the parallel circuit with the series circuit. [1]

....

(ii) Explain how the voltmeter readings in the two circuits compare. [2]

....

....

....

(iii) Explain how the power dissipated in resistor R_1 in each circuit compares. [2]

....

....

....

(b) Jamie used identical 2.25 Ω resistors in both circuits.

(i) Calculate the power dissipated by R_1 in the parallel circuit. [4]

power dissipated = ... W

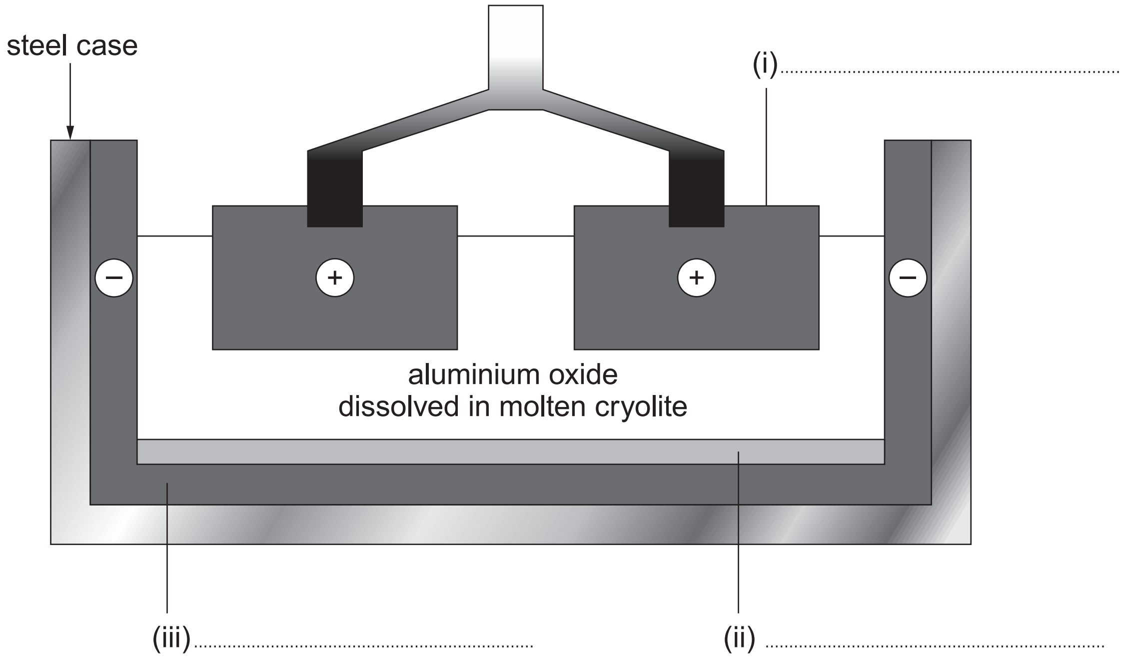
(ii) Use your answer above to calculate the power dissipated in resistor R_1 in the series circuit. [2]

power dissipated = ... W

13

Examiner only

7. The diagram below shows the extraction of aluminium from aluminium oxide by electrolysis.



(a) Complete the labelling of the diagram above. [3]

(b) Complete the balanced **symbol** equation for this process. [3]

aluminium oxide ⟶ aluminium + oxygen

$\square\ Al_2O_3 \longrightarrow \square$ .... + $\square$ ....

(c) Explain, in terms of electrons, what happens at each of the electrodes. [3]

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Examiner only

(d) Over the last 10 years the mass of aluminium recycled in Wales has increased.

(i) Suggest **two** economic benefits of recycling aluminium. [2]

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....

....

....

(ii) Explain **one** environmental benefit of recycling aluminium. [2]

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....

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13

END OF PAPER

THE PERIODIC TABLE

1	2	Group										3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number